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Sodium Iodide/*tert*-Butyl(dimethyl)silyl Chloride-Induced Isomerization of 2,3-Allenols to 2(*E*)-Enals

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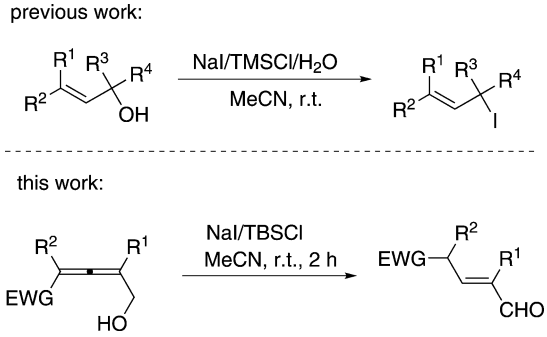
Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201300835.

Abstract: Through an unexpected C=C bond migration, 2(*E*)-enals were efficiently prepared highly stereoselectively in moderate to good yields *via* the sodium iodide/*tert*-butyl(dimethyl)silyl chloride (NaI/TBSCl)-mediated reaction of the easily available 2,3-allenols. Based on a careful mechanistic study, including control experiments and deuterium-labeling experiments, an electron-withdrawing substituent group in the 4 position has been proven to be crucial and a possible mechanism has been proposed.

Keywords: allenols; electrophilic addition; migration; oxidation; α,β-unsaturated aldehydes

Introduction

α,β-Unsaturated aldehydes are an important class of chemical compounds because of the wide existence of this unit in nature, the extensive use in food, cosmetic and pharmaceutical industries,[1] and for being valuable intermediates in organic synthesis.[2] Therefore, there have been many practical methods for the synthesis of the this type of compounds.[3–9] On the other hand, as an equivalent to TMSI, TMSCl/NaI reagent has been widely used in synthetic organic chemistry.[10,11] In 1989, Y. Ishii et al. reported a facile iodination of allylic alcohols with the NaI/TMSCl/H_2O forming allylic iodides (Scheme 1).[11b] However, when we applied this set of reagents to 2,3-allenols with an electron-withdrawing group on the allene unit,[12,13] 2(*E*)-enals substituted with an electron-withdrawing group at the 4 position were unexpectedly formed upon the treatment with NaI and TBSCl (Scheme 1). Herein, we wish to report the details of this transformation.



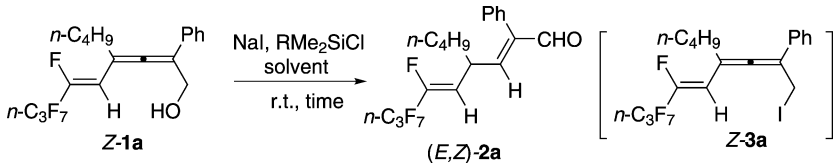
Scheme 1.

Results and Discussion

At the beginning of our work, the reaction of perfluoroalkyl-containing 2,3-allenol **1a**, prepared following the known procedure,[14] with NaI and TMSCl was carried out in MeCN at room temperature with the purpose of preparing the corresponding allenyl iodide. Surprisingly, instead of forming the expected allenyl iodide product *Z*-**3a**, a new product was formed unexpectedly. Interestingly, the $^1H/^{13}C$ NMR and IR analysis of this product indicated the presence of an aldehyde functionality. Further study of the mass spectra led us to the conclusion that this product should be 4-butyl-6,7,7,8,8,9,9,9-octafluoro-2-phenyl-2(*E*),5(*Z*)-nonadienal **2a** (Table 1). The configuration of the double bond in the 2-position was determined by a NOESY analysis.[15] Having established the identity of **2a**, the solvent effect was then examined with 2.0 equiv. each of NaI and TMSCl at room temperature to optimize the reaction yield (entries 1–6, Table 1): toluene gave a similar result with benzene (entries 1 and 4, Table 1); using THF as solvent, the reaction was messy (entry 3, Table 1); MeCN re-

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Table 1. Optimization of reaction conditions for the reaction of **1a** with NaI and chlorosilanes.[a]



Entry	Solvent	R	Time [h]	Yield [%][b] of (*E*,*Z*)-**2a**	(*E*,*Z*)/(*Z*,*Z*)
1	toluene	Me	13	44	98/2
2	CH_2Cl_2	Me	2	38	99/1
3	THF	Me	2	complicated	–
4	benzene	Me	2	48	98/2
5	DCE	Me	2	53	98/2
6	MeCN	Me	2	61	98/2
7	MeCN	–[c]	2	0[d]	–
8[e]	MeCN	Me	2	complicated	–
9[f]	MeCN	Me	1.5	58	98/2
10	MeCN	*t*-Bu[g]	2	66	98/2
11	MeCN	*t*-Bu[h]	2	62	98/2
12[g,i]	MeCN	*t*-Bu	2	59	98/2
13[g,j]	MeCN	*t*-Bu	2	72 (70)	98/2

[a] The reaction was carried out using 0.1 mmol or 0.15 mmol of **1a**, 2.0 equiv. of NaI, and 2.0 equiv. of RMe_2SiCl in 2 mL of solvent.
[b] Determined by ^1H NMR analysis using mesitylene as the internal standard.
[c] RMe_2SiCl was not used.
[d] 97% of **1a** was recovered.
[e] NaI was not used.
[f] 2.0 equiv. of KI were used.
[g] 1.5 equiv. of *t*-$BuMe_2SiCl$ were used.
[h] 1.2 equiv of *t*-$BuMe_2SiCl$ were used.
[i] 1.5 equiv. of NaI were used.
[j] The concentration of **1a** in MeCN was 0.1 M.

mained the best with a 61% yield (entry 6, Table 1). In addition, this reaction did not proceed in the absence of either NaI or TMSCl (entries 7 and 8, Table 1). The yield of **2a** could be improved to 66% when 1.5 equiv. of TBSCl were used, and further increased to 72% yield when the concentration of **1a** in MeCN was adjusted to 0.1 M (entry 13, Table 1). Thus, we defined the experimental protocol for the migrative oxidation of 4-electron-withdrawing group-substituted 2,3-allenols with 2.0 equiv. NaI and 1.5 equiv. TBSCl in MeCN at room temperature as the standard reaction conditions for further study.

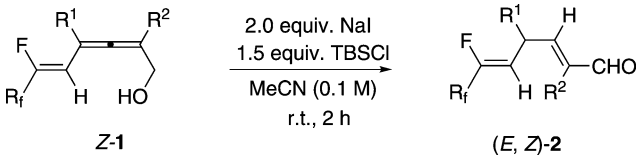
With the optimized conditions in hand, the scope of this reaction was studied, and the typical results are summarized in Table 2. Differently substituted 6-fluoro-6-perfluoroalkyl-2,3,5(*Z*)-trien-1-ols **1a–1n** were successfully applied in this reaction to afford 6-fluoro-6-perfluoroalkyl-2(*E*),5(*Z*)-dienals **2a–2n** in moderate to good yields with high stereoselectivity. The substituent R^1 could be alkyl groups, such as *n*-butyl (entries 1–10, Table 2) or *n*-hexyl group (entries 11–13, Table 2); the R_f group could be *n*-C_3F_7 (entries 1–8 and 11–13, Table 2), *n*-C_5F_{11} (entries 9 and 10, Table 2), or CF_3 (entry 14, Table 2); the R^2 group may be phenyl (entries 1, 9 and 11, Table 2), substituted phenyl (entries 2–6, 10, 12 and 13, Table 2) or a thienyl group (entry 7, Table 2). The reaction of substrates with R^2 being an electron-donating phenyl group (entries 2–4 and 12, Table 2) gave better results than that of the phenyl (entries 1, 9, 10 and 14, Table 2) or those with an electron-withdrawing group on the phenyl ring (entries 5, 6, 10 and 13, Table 2). Alkyl groups may also be introduced as the R^2 group (entry 8, Table 2), although the yield was lower. In order to further clarify the structure of this compound, we attempted the condensation reaction of **2n** with 2,4-dinitrophenylhydrazine to afford a solid product (*E*,*Z*)-**4a**, whose X-ray single crystal diffraction study definitely showed that the product **2n** is 6,6,6,7-tetrafluoro-4-butyl-2-phenyl-2-heptenal in the *E,Z*-configuration (Figure 1).[16]

In order to further extend the scope, 4-methyl-2-phenyl-2,3-pentadienol **1p** and 5,5,5-trifluoro-4-methyl-2-phenyl-2,3-pentadienol **1q** were prepared[17] and submitted to the standard conditions, respectively. Interestingly, the reaction of 4-trifluoromethyl-substituted substrate **1q** led to the similar aldehyde product *E*-**2q** in 55% yield while the reaction of **1p** was com-

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Table 2. Highly stereoselective synthesis of α,β-unsaturated aldehydes by isomerization of fluorine-containing allenols.



Z-**1** (*E*, *Z*)-**2**

Entry	$R^1/R^2/R_f$	Isolated yield [%] of (*E*,*Z*)-**2**	(*E*,*Z*)/(*Z*,*Z*)[a]
1	*n*-C_4H_9/Ph/*n*-C_3F_7 [(*Z*)-**1a**]	70 [(*E*,*Z*)-**2a**]	98/2
2	*n*-C_4H_9/4-$MeOC_6H_4$/*n*-C_3F_7 [(*Z*)-**1b**]	80 [(*E*,*Z*)-**2b**]	98/2
3	*n*-C_4H_9/4-MeC_6H_4/*n*-C_3F_7 [(*Z*)-**1c**]	78 [(*E*,*Z*)-**2c**]	98/2
4	*n*-C_4H_9/3-MeC_6H_4/*n*-C_3F_7 [(*Z*)-**1d**]	68 [(*E*,*Z*)-**2d**]	98/2
5	*n*-C_4H_9/4-FC_6H_4/*n*-C_3F_7 [(*Z*)-**1e**]	68 [(*E*,*Z*)-**2e**]	98/2
6	*n*-C_4H_9/4-BrC_6H_4/*n*-C_3F_7 [(*Z*)-**1f**]	53 [(*E*,*Z*)-**2f**]	98/2
7	*n*-C_4H_9/2-thienyl/*n*-C_3F_7 [(*Z*)-**1g**]	60 [(*E*,*Z*)-**2g**]	95/5
8	*n*-C_4H_9/*n*-C_4H_9/*n*-C_3F_7 [(*Z*)-**1h**]	32 [(*E*,*Z*)-**2h**]	99/1
9	*n*-C_4H_9/Ph/*n*-C_5F_{11} [(*Z*)-**1i**]	60 [(*E*,*Z*)-**2i**]	98/2
10	*n*-C_4H_9/4-FC_6H_4/*n*-C_5F_{11} [(*Z*)-**1j**]	66 [(*E*,*Z*)-**2j**]	98/2
11	*n*-C_6H_{13}/Ph/*n*-C_3F_7 [(*Z*)-**1k**]	68 [(*E*,*Z*)-**2k**]	98/2
12	*n*-C_6H_{13}/4-MeC_6H_4/*n*-C_3F_7 [(*Z*)-**1l**]	80 [(*E*,*Z*)-**2l**]	98/2
13	*n*-C_6H_{13}/4-FC_6H_4/*n*-C_3F_7 [(*Z*)-**1m**]	69 [(*E*,*Z*)-**2m**]	98/2
14	*n*-C_4H_9/Ph/CF_3 [(*Z*)-**1n**]	62 [(*E*,*Z*)-**2n**]	98/2

[a] The (*E*,*Z*)/(*Z*,*Z*) ratio of **2** was determined by ^{1}H NMR analysis of the crude product.

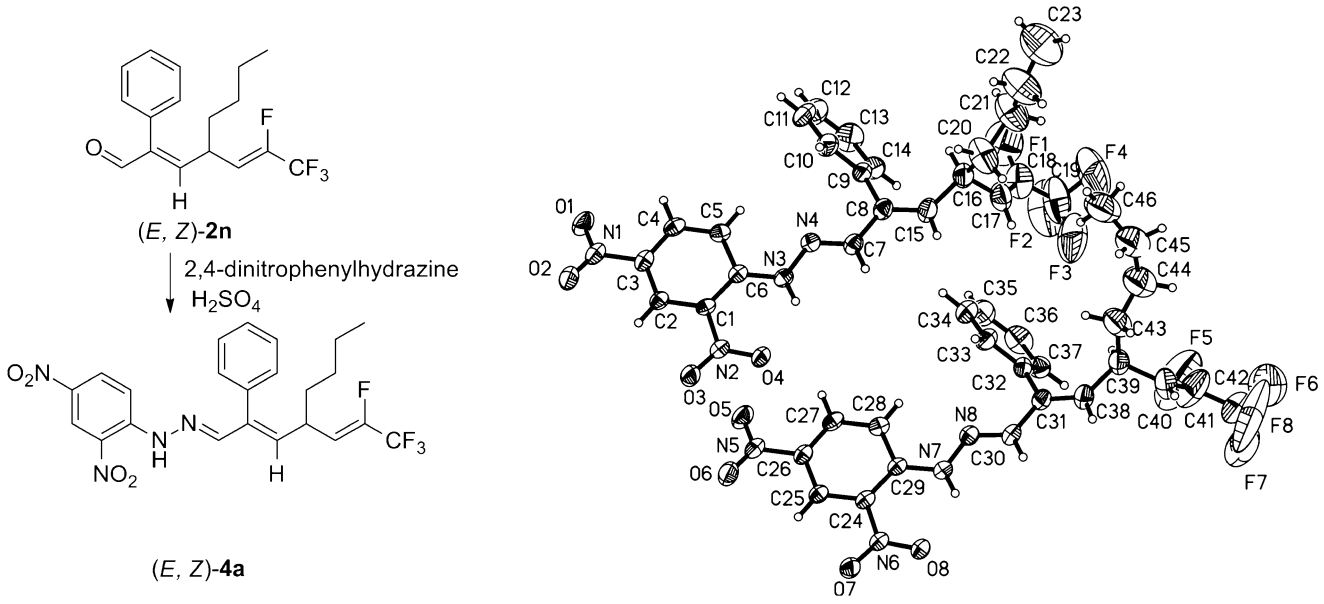


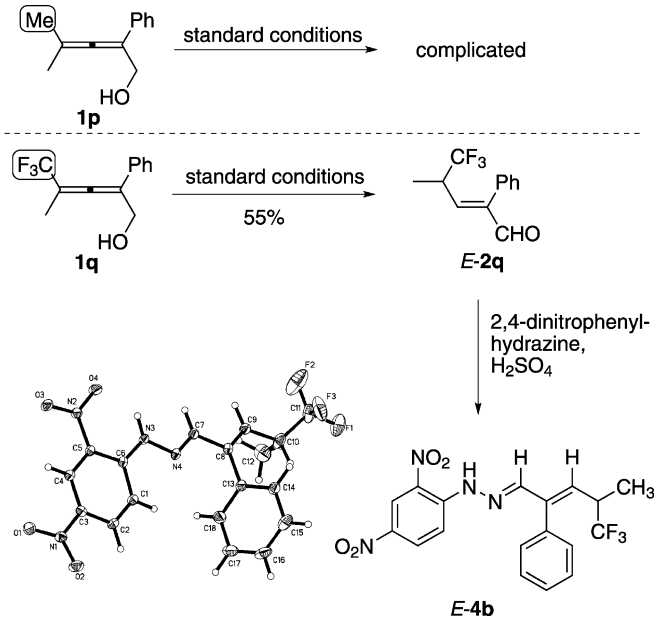
Figure 1. Formation and ORTEP representation of (*E*,*Z*)-**4a**.

plicated, which indicated that an electron-withdrawing group at the 4-position of allenols is essential for this reaction (upper *vs.* middle parts, Scheme 2). The condensation reaction of *E*-**2q** with 2,4-dinitrophenylhydrazine afforded a solid product *E*-**4b**, whose X-ray single crystal diffraction study definitely clarified the structure of *E*-**2q** (lower part, Scheme 2).[18]

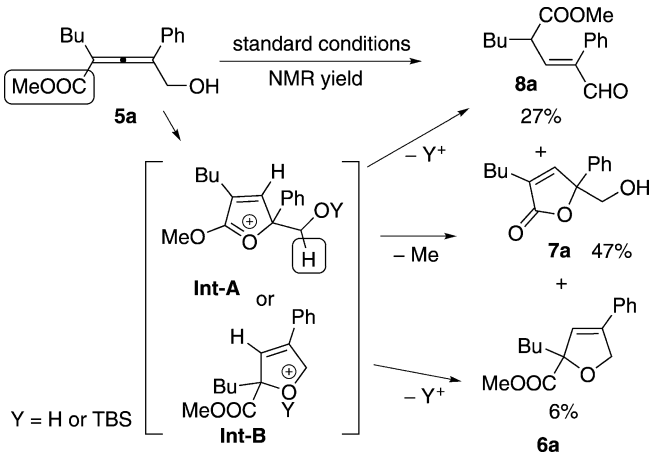
This electronic effect of the electron-withdrawing group was further confirmed by the fact that the reaction of **5a** with the electron-withdrawing methoxycarbonyl group also afforded the expected α,β-unsaturated aldehyde **8a** in 27% yield, although the major product is the lactonization product **7a** involving the ester functionality as expected (47% yield).[13k] In addition to these two products, only 6% of cyclization product involving the hydroxy group,[13k] **6a**, was observed in this reaction based on the ^{1}H NMR analysis of the crude reaction mixture (Scheme 3). These ex-

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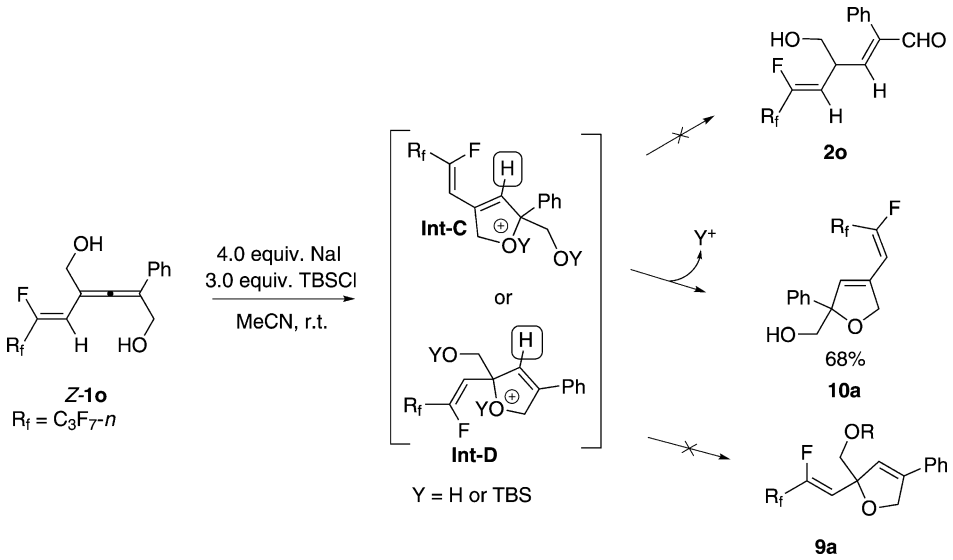
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Scheme 2.



Scheme 3.



Scheme 4.

perimental results indicate the presence of cationic intermediates **Int-A**[13k] and **Int-B**[13k] forming from electrophilic addition of **5a** with H^+. Due to the destabilization effect of the electron-withdrawing methoxycarbonyl group, **Int-A** was prefered over **Int-B**, affording **8a** and **7a** as the major products.

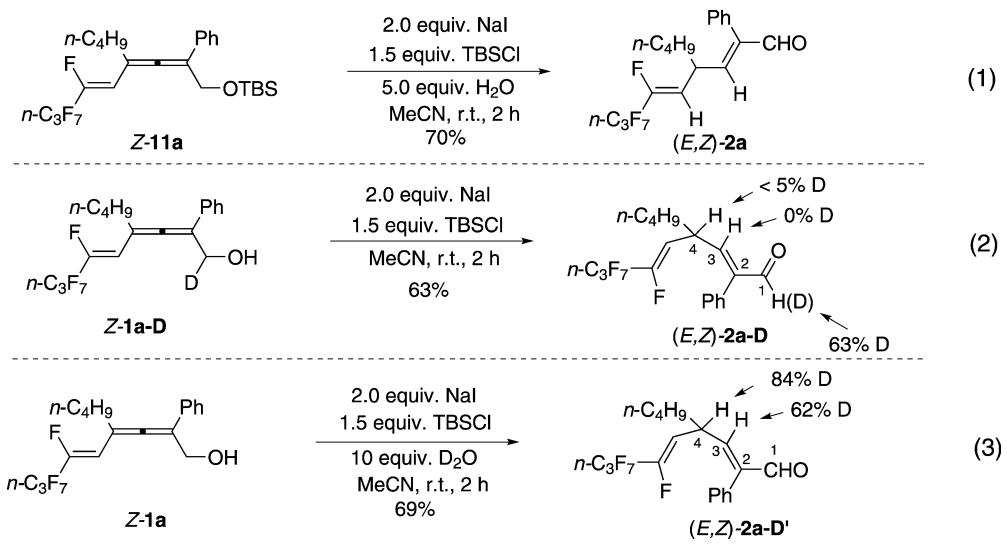
A similar cyclization was observed with the reaction of *Z*-**1o**, in which there is an extra hydroxymethyl group at the 4-position: 2,5-dihydrofuran **10a** was formed as the major product, while the expected product **2o** and the other cyclization product **9a** were not detected in the crude product. These experimental results may nicely explain that H^+ is indeed the active species under the current reaction conditions affording the product **10a** *via* highly efficient H^+-mediated electrophilic cyclic etherification of *Z*-**1o** (Scheme 4).

Then the corresponding allenol TBS ether *Z*-**11a** was synthesized and treated with NaI and TBSCl in the presence of 5 equiv. of water. After 2 h, the expected aldehyde product **2a** was formed in 70% isolated yield [Eq. (1), Scheme 5]. Furthermore, a deuterium-labeling experiment was performed: the reaction of **1a-D** under the standard condition affording (*E*,*Z*)-**2a-D** with 0% and less than 5% D at the C-3 and C-4 positions, respectively, while 63% D was observed at C-1, indicating an isotopic effect [Eq. (2), Scheme 5]. The reaction of *Z*-**1a** was performed in the presence of 10 equiv. of D_2O affording the (*E*,*Z*)-**2a-D′** with 62% and 84% D incorporated into the C-3 and C-4 positions, respectively [Eq. (3), Scheme 5].

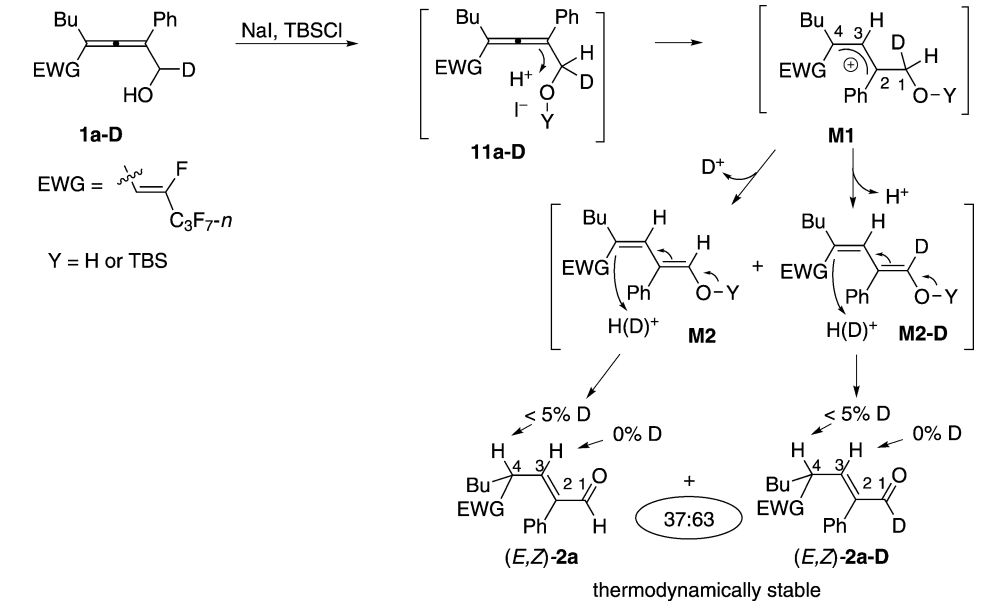
With the evidence above, we proposed a mechanism for this transformation (Scheme 6). The reaction of fluorine-containing 2,3-allenol **1a-D** with NaI and TBSCl would form the allenol TBS ether **11a-D** and

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Scheme 5.



Scheme 6.

generate HI *in situ*. The electrophilic addition of the allene unit intermediate **11a-D** with H^+ would form π-allylic cation **M1**. Due to the destabilization effect of the electron-withdrawing fluorine-containing substituent at the 4-position of the substrate, the cation center was inclined to the C-2 position. H^+- or D^+-elimination of **M1** would form conjugated trienol silyl ether **M2** or **M2-D**, respectively, which upon reaction with H^+ or D^+ would afford the thermodynamically more stable product (*E*,*Z*)-**2a** and (*E*,*Z*)-**2a-D**. The rapid proton-exchange and kinetic isotopic effect would lead to the low D-incorporation at the C-4 position and 63% D-incorporation at the C-1 position of the product.

Conclusions

In summary, we have observed a unique C=C migration of 4-electron-withdrawing group-substituted 2,3-allenols, which provides an efficient synthesis of 4-electron-withdrawing group-substituted 2(*E*)-enals. Due to the easy availability of the starting allenols,[14] and the highly functionalized nature of the products, this method would be useful in organic synthesis. Based on a careful mechanistic study, a possible mechanism has been proposed. Further studies in this area are being carried out in our laboratory.

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Experimental Section

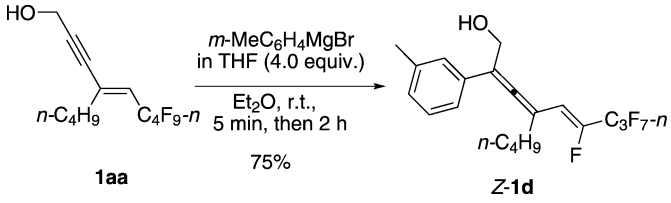
Materials

THF and Et_2O were distilled from Na/benzophenone before use. MeCN and DCM were distilled from CaH_2 before use. PhMgCl (1.8 M solution in THF) was purchased from Acros. Other Grignard reagents were prepared from the corresponding aryl halides and magnesium. Other commercially available chemicals were purchased and used without additional purification unless noted otherwise. ^{1}H NMR spectra were recorded on a Bruker-300 MHz spectrometer and ^{13}C NMR spectra were recorded at 75 MHz. All ^{1}H NMR experiments were measured with tetramethylsilane (0 ppm) in $CDCl_3$ as the internal reference and ^{13}C NMR experiments were measured in relative to the signal of $CDCl_3$ (77.0 ppm) in $CDCl_3$. Due to the coupling of multiple fluorines, ^{13}C NMR signals of perfluoroalkyl group are extremely weak, and ^{19}F NMR experiments were undecoupled and measured relative to the signal of residual $CFCl_3$ (0 ppm) in $CDCl_3$. The coupling constant in the ^{1}H NMR spectra larger than 30 Hz is the H,F coupling.

Preparation of Starting Allenols

Compounds **1a–1c**, **1e**, **1h–1i**, **1k–1m** as well as new compounds **1d**, **1f, 1g**, **1j**, **1n** and **1o** were prepared according to the literature procedures.[14]

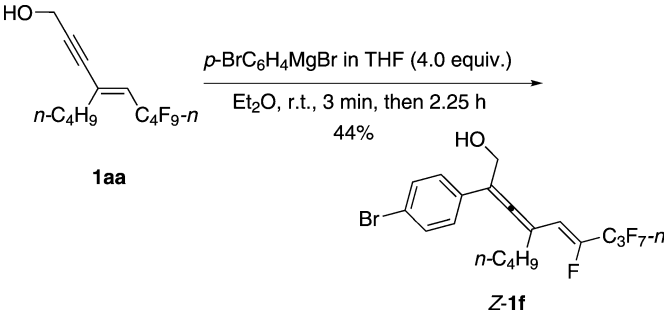
Typical Procedure 1: 4-(*n*-Butyl)-6,7,7,8,8,9,9,9-octafluoro-2-(3-methylphenyl)nona-2,3,5(*Z*)-trienol (*Z*-1d)



To a dry Schlenk tube containing **1aa** (1.0689 g, 3.0 mmol) and 7.5 mL of anhydrous Et_2O was added *m*-MeC_6H_4MgBr (2.0 M in THF, 6.0 mL, 12.0 mmol) dropwise by syringe under a nitrogen atmosphere at room temperature within 5 min. Then the resulting reaction mixture was stirred at room temperature for 2 h. After quenching with an aqueous solution of saturated ammonium chloride (5 mL) and water (15 mL), the resulting mixture was extracted with diethyl ether (30 mL × 3), washed with brine once, and dried over anhydrous Na_2SO_4. Filtration, evaporation and column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 30/1 to 20/1) afforded *Z*-**1d** as an oil; yield: 0.9684 g (75%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.30–7.22 (m, 1H, Ar-H), 7.22–7.14 (m, 2H, Ar-H), 7.13–7.05 (m, 1H, Ar-H), 5.92 (d, J = 34.2 Hz, 1H, CH=), 4.59 (s, 2H, CH_2O), 2.45–2.33 (m, 5H, CH_2 and CH_3), 1.69 (s, 1H, OH), 1.55–1.30 (m, 4H, $CH_2 \times 2$), 0.91 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$):δ = 207.4 (d, J = 5.2 Hz), 144.8 (dt, J_1 = 266.9 Hz, J_2 = 28.2 Hz), 138.4, 133.3, 128.7, 128.6, 127.1, 123.5, 122.1–111.8 (m), 108.5, 103.4 (d, J = 2.9 Hz), 61.8, 31.7 (d, J = 2.8 Hz), 30.3, 22.2, 21.5, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.0 ~(−118.2) (m, 2F), −126.9 ~(−127.2) (m, 1F), −127.2 ~(−127.4) (m, 2F); IR (neat): ν = 3374, 2960, 2932, 2864, 1933, 1681, 1604, 1490, 1467, 1360, 1267, 1231, 1187, 1152, 1119, 1058, 1033, 951, 919 cm^{-1}; MS (EI, 70 eV): m/z (%) = 428 (M^+, 1.50), 368 (100); HR-MS: m/z = 428.1391, calcd. for $C_{20}H_{20}F_8O$ (M^+): 428.1386.

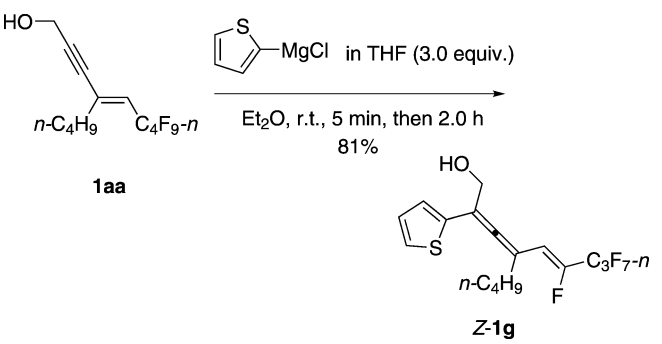
The following compounds were prepared according to Procedure 1.

4-(*n*-Butyl)-6,7,7,8,8,9,9,9-octafluoro-2-(4-bromophenyl)-nona-2,3,5(*Z*)-trienol (*Z*-1f): The reaction of **1aa** (1.0684 g,



3.0 mmol), Et_2O (8 mL), and *p*-BrC_6H_4MgBr (1.6 M in THF, 7.5 mL, 12 mmol) afforded *Z*-**1f** (eluent: petroleum ether/DCM = 3/1) as an oil; yield: 0.6448 g (44%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.54–7.43 (m, 2H, Ar-H), 7.34–7.20 (m, 2H, Ar-H), 5.90 (d, J = 34.2 Hz, 1H, CH=), 4.58 (d, J = 5.7 Hz, 2H, CH_2O), 2.38 (t, J = 7.2 Hz, 2H, CH_2), 1.74–1.62 (m, 1H, OH), 1.56–1.28 (m, 4H, $CH_2 \times 2$), 0.90 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 207.3 (d, J = 5.0 Hz), 145.2 (dt, J_1 = 267.5 Hz, J_2 = 28.1 Hz), 132.5, 131.8, 128.0, 121.6, 111.7–111.4 (m), 107.6, 103.5 (d, J = 2.5 Hz), 61.7, 31.8 (d, J = 2.9 Hz), 30.3, 22.2, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.0 ~(−118.3) (m, 2F), −125.8 ~(−126.2) (m, 1F), −127.1 ~(−127.4) (m, 2F); IR (neat); ν = 3357, 2961, 2933, 2875, 1934, 1682, 1489, 1467, 1361, 1229, 1185, 1153, 1120, 1076, 1038, 1009 cm^{-1}; MS (EI, 70 eV): m/z (%) = 494 (M^+ (^{81}Br), 1.90), 492 (M^+ (^{79}Br), 1.78), 341 (100); HR-MS: m/z = 492.0329, calcd. for $C_{19}H_{17}{}^{79}BrF_8O$ (M^+ (^{79}Br)): 492.0335.

4-(*n*-Butyl)-6,7,7,8,8,9,9,9-octafluoro-2-(2-thiophenyl)no-na-2,3,5(*Z*)-trienol (*Z*-1g): The reaction of **1aa** (1.0655 g,

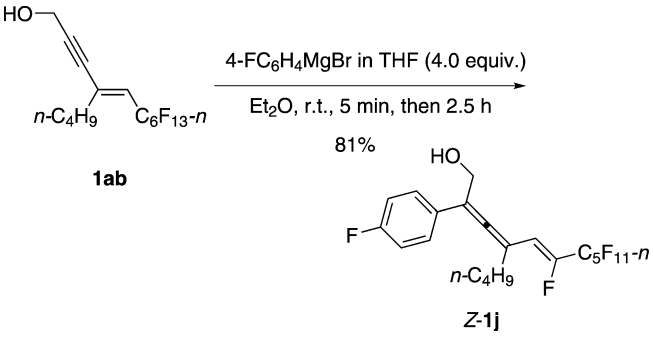


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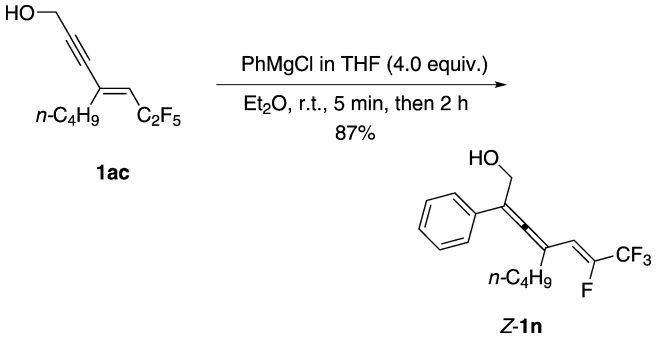
3.0 mmol), Et_2O (7.5 mL), and 2-thiophenylMgBr (2.0 M in THF, 4.5 mL, 9.0 mmol) afforded *Z*-**1g** as an oil (eluent: petroleum ether/ethyl acetate = 40/1 to 20/1); yield 1.0191 g (81%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.30–7.20 (m, 1H, Ar-H), 7.10–6.96 (m, 2H, Ar-H), 5.90 (d, J = 33.9 Hz, 1H, CH=), 4.55 (d, J = 5.4 Hz, 2H, CH_2O), 2.38 (t, J = 7.5 Hz, 2H, CH_2), 1.76 (t, J = 5.7 Hz, 1H, OH), 1.58–1.30 (m, 4H, $CH_2\times2$), 0.91 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 206.6 (d, J = 5.4 Hz), 145.1 (dt, J_1 = 267.3 Hz, J_2 = 28.4 Hz), 137.3, 127.6, 125.4, 124.2, 111.8–111.4 (m), 104.3, 103.7 (d, J = 2.9 Hz), 62.4, 32.0 (d, J = 2.4 Hz), 30.2, 22.2, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.1 ~(−118.3) (m, 2F), −125.9 ~(−126.3) (m, 1F), −127.2 ~(−127.4) (m, 2F); IR (neat): ν = 3374, 2961, 2933, 2875, 1933, 1681, 1467, 1360, 1228, 1187, 1154, 1120, 1034, 957, 913 cm^{-1}; MS (EI, 70 eV): *m/z* (%) = 420 (M^+, 2.35), 347 (100); HR-MS: *m/z* = 420.0791, calcd. for $C_{17}H_{16}F_8OS$ (M^+): 420.0794.

4-(*n*-Butyl)-6,7,7,8,8,9,9,9-octafluoro-2-(2-thiophenyl)nona-2,3,5(*Z*)-trienol (*Z*-1j): The reaction of **1ab** (1.1093 g, 2.5 mmol), Et_2O (6.5 mL), and *p*-FC_6H_4MgBr (2.0 M in THF, 5.0 mL, 10 mmol) afforded *Z*-**1j** as an oil (eluent: petroleum ether/ethyl acetate = 100/1 to 60/1 to 40/1 to 30/1); yield: 1.0700 g (81%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.42–7.32 (m, 2H, Ar-H), 7.11–6.98 (m, 2H, Ar-H), 5.91 (d, J = 34.2 Hz, 1H, CH=), 4.58 (d, J = 6.0 Hz, 2H, CH_2O), 2.38 (t, J = 7.4 Hz, 2H, CH_2), 1.71 (t, J = 6.2 Hz, 1H, OH), 1.60–1.30 (m, 4H, $CH_2\times2$), 0.90 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 207.2 (d, J = 3.3 Hz), 163.9, 160.6, 145.2 (dt, J_1 = 267.2 Hz, J_2 = 28.3 Hz), 129.4 (d, J = 2.9 Hz), 128.1 (d, J = 7.7 Hz), 115.8 (d, J = 21.7 Hz), 112.0–111.7 (m), 107.5, 103.4 (d, J = 2.9 Hz), 61.9, 31.8 (d, J = 3.0 Hz), 30.3, 22.2, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.7 ~(−80.9) (m, 3F), −114.1 ~(−114.3) (m, 1F), −117.0 ~(−117.3) (m, 2F), −122.7 ~(−123.2) (m, 4F), −126.1 ~(−126.5) (m, 3F); IR (neat): ν = 3342, 2961, 2933, 2875, 1933, 1682, 1603, 1510, 1468, 1364, 1237, 1203, 1143, 1110, 1047 cm^{-1}; MS (EI, 70 eV): *m/z* (%) = 532 (M^+, 1.62), 109 (100); HR-MS: *m/z* = 532.1068, calcd. for $C_{21}H_{17}F_{13}O$ (M^+): 532.1072.



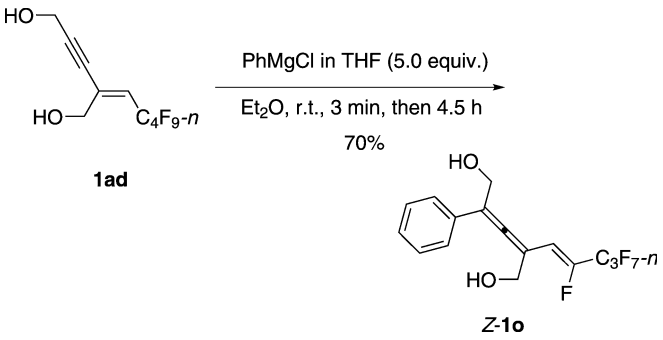
1ab Z-1j

4-(*n*-Butyl)-2-phenyl-6,7,7,7-tetrafluorohepta-2,3,5(*Z*)-trienol (*Z*-1n): The reaction of **1ac** (440.4 mg, 1.72 mmol), Et_2O (6 mL), and PhMgCl (2.0 M in THF, 3.6 mL, 7.2 mmol) afforded *Z*-**1n** as an oil (eluent: petroleum ether/ethyl acetate = 20/1 to 10/1); yield: 469.0 mg (87%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.46–7.18 (m, 5H, Ar-H), 5.90 (d, J = 34.8 Hz, 1H, CH=), 4.58 (d, J = 3.9 Hz, 2H, CH_2O), 2.37 (t, J = 7.4 Hz, 2H, CH_2), 2.02–1.90 (m, 1H, OH), 1.60–1.28 (m, 4H, $CH_2\times2$), 0.90 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 207.2 (d, J = 5.5 Hz), 144.8 (dq, J_1 = 265.1 Hz, J_2 = 38.0 Hz), 133.4, 128.7, 127.6, 126.4, 118.6 (qd, J_1 = 270.3 Hz, J_2 = 40.7 Hz), 109.5–109.1 (m), 108.3, 103.2 (d, J = 3.1 Hz), 61.6, 31.6 (d, J = 3.2 Hz), 30.3, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −72.5 (d, J = 12.7 Hz, 3F), −131.5 (dq, J_1 = 33.7 Hz, J_2 = 12.0 Hz, 1F); IR (neat): ν = 3359, 2960, 2932, 2874, 1934, 1689, 1598, 1496, 1467, 1359, 1307, 1288, 1188, 1141, 1047, 1026 cm^{-1}; MS (EI, 70 eV): *m/z* (%) = 314 (M^+, 1.38), 254 (100); HR-MS: *m/z* = 314.1307, calcd. for $C_{17}H_{18}F_4O$ (M^+): 314.1294.



1ac Z-1n

2-[2,3,3,4,4,5,5,5-(*Z*)-Octafluoropentenyl]-4-phenylpenta-2,3-diene-1,5-diol (*Z*-1o): The reaction of **1ad** (64.9 mg, 0.2 mmol), Et_2O (0.5 mL), and PhMgCl (1.8 M in THF, 0.56 mL, 1.0 mmol) afforded *Z*-**1o** as an oil (eluent: petroleum ether/ethyl acetate = 3/1 for the first round and DCM/ethyl acetate = 5/1 to 4/1 for the second round); yield: 53.3 mg (70%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.22 (m, 5H, Ar-H), 5.97 (d, J = 36.0 Hz, 1H, CH=), 5.59 (d, J = 13.2 Hz, 1H, CH_2O), 5.50 (d, J = 13.2 Hz, 1H, CH_2O), 4.36 (s, 2H, CH_2O), 3.90 (bs, 2H, OH × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 206.6 (d, J = 5.6 Hz), 145.2 (dt, J_1 = 266.3 Hz, J_2 = 28.4 Hz), 132.8, 128.8, 128.1, 126.4, 111.3, 109.8–109.4 (m), 104.5 (d, J = 4.1 Hz), 61.5 (d, J = 5.0 Hz), 61.1; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.9 (t, J = 8.3 Hz, 3F), −118.0 ~(−118.2) (m, 2F), −125.3 ~(−125.7) (m, 1F), −127.2 (d, J = 2.5 Hz, 2F); IR (neat): ν = 3332, 2935, 2880, 1939, 1683, 1496, 1453, 1360, 1228, 1187, 1155, 1120, 1053, 1029 cm^{-1}; MS (ESI): *m/z* (%) = 411 (M^++Na, 100); HR-MS: *m/z* = 411.0617, calcd. for $C_{16}H_{12}F_8NaO_2$ (M^+): 411.0602.

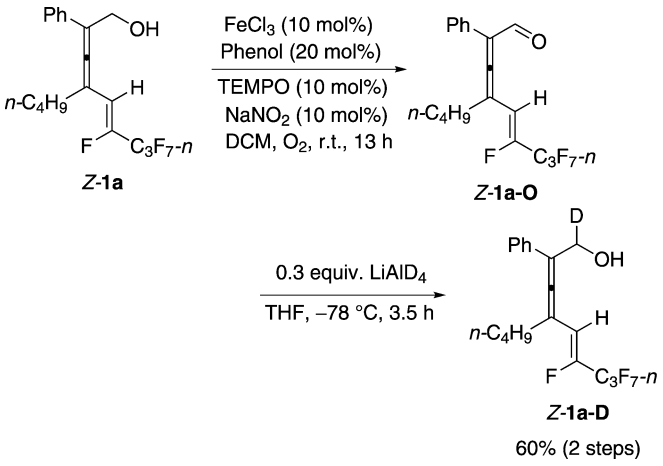


1ad Z-1o

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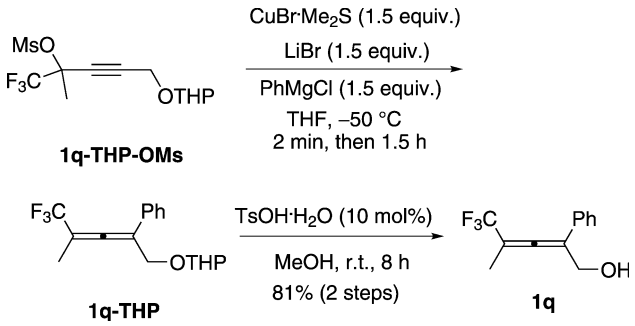
4-(*n*-Butyl)-1-deuterio-6,7,7,8,8,9,9,9-octafluoro-2-phenylnona-2,3,5(*Z*)-trienol (*Z*-1a-D)



Compound *Z*-**1a-D** was prepared by oxidation and reduction of *Z*-**1a**.[19] To a dried Schlenk tube equipped with a Teflon-coated magnetic stirring bar were added $FeCl_3$ (24.6 mg, 0.15 mmol), TEMPO (23.8 mg, 0.15 mmol), $NaNO_2$ (10.2 mg, 0.15 mmol), phenol (29.2 mg, 0.3 mmol), and DCM (2 mL) sequentially at room temperature under an oxygen atmosphere (1 atm). After stirring for 10 min, *Z*-**1a** (621.6 mg, 1.5 mmol) and DCM (4 mL) were added. The resulting mixture was stirred for 13 h, filtrated through a short column of silica gel (eluent: Et_2O), purified by flash column chromatography on silica gel (eluent: petroleum ether/ethyl ether = 80/1) to afford *Z*-**1a-O**, which was used for the reduction reaction directly.

To a dried Schlenk tube equipped with a Teflon-coated magnetic stirring bar were added *Z*-**1a-O** and THF (4 mL) at room temperature under a nitrogen atmosphere. The tube was then submerged in a cooling bath at −78 °C. After stirring for 5 min, $LiAlD_4$ (19.0 mg, 0.45 mmol) and THF (2 mL) were added. After 3.5 h, the reaction was quenched with a saturated aqueous solution of NH_4Cl (1 mL) at −78 °C, and the aqueous layer was extracted with Et_2O (25 mL × 3). The combined organic layer was washed with brine, dried over anhydrous Na_2SO_4. Filtration, evaporation, and silica gel flash column chromatography (eluent: petroleum ether/ethyl acetate = 20/1 to 10/1) afforded the product *Z*-**1a-D** as an oil; yield: 375.8 mg (60%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.46–7.24 (m, 5H, Ar-H), 5.92 (d, J = 34.2 Hz, 1H, CH=), 4.65–4.55 (m, 1H, CHO), 2.39 (t, J = 7.4 Hz, 2H, CH_2), 1.62 (s, 1H, OH), 1.57–1.30 (m, 4H, $CH_2 \times 2$), 0.90 (t, J = 7.1 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 207.4 (d, J = 4.9 Hz), 144.9 (dt, J_1 = 268.7 Hz, J_2 = 28.6 Hz), 133.4, 128.8, 127.7, 126.4, 112.0–111.7 (m), 108.4, 103.5 (d, J = 2.7 Hz), 61.4 (t, J = 22.6 Hz), 31.8 (d, J = 3.8 Hz), 30.4, 22.3, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 (t, J = 9.4 Hz, 3F), −118.0 ~ (−118.2) (m, 2F), −126.6 ~ (−127.0) (m, 1F), −127.2 ~ (−127.4) (m, 2F); IR (neat): ν = 3340, 2961, 2933, 1933, 1682, 1599, 1496, 1453, 1361, 1230, 1188, 1153, 1120, 1059, 1032 cm^{-1}; MS (EI, 70 eV): m/z (%) = 415 (M^+, 1.45), 355 (100); HR-MS: m/z = 415.1283, calcd. for $C_{19}H_{17}DF_8O$ (M^+): 415.1293.

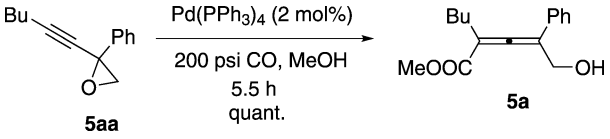
4-Methyl-2-phenyl-5,5,5-trifluoro-2,3-pentadienol (1q)



Compound **1q** was prepared according to the literature procedure.[20] To a dried Schlenk tube equipped with a Teflon-coated magnetic stirring bar were added LiBr (52.0 mg, 0.6 mmol), $CuBr{\cdot}Me_2S$ (124.3 mg, 0.6 mmol), THF (2 mL), **1q-THP-OMs** (133.3 mg, 0.4 mmol), and THF (2 mL) sequentially at −50 °C under a nitrogen atmosphere. Then PhMgCl (1.8 M in THF, 0.33 mL, 0.6 mmol) was added dropwise using the syringe in 2 min at −50 °C. After 1.5 h, the reaction was quenched with a saturated aqueous solution of NH_4Cl (2 mL) at −50 °C, and the aqueous layer was extracted with Et_2O (15 mL × 3). The combined organic layer was dried over anhydrous Na_2SO_4. Filtration, evaporation, and silica gel flash column chromatography (eluent: petroleum ether/ethyl acetate = 100/1 to 30/1) afforded the product **1q-THP**, which was used in the next step directly.

To a dried Schlenk tube equipped with a Teflon-coated magnetic stirring bar were added **1q-THP**, MeOH (3 mL), and $TsOH{\cdot}H_2O$ (7.8 mg, 0.04 mmol) sequentially at room temperature. After 8.0 h, the reaction mixture was evaporated, and silica gel flash column chromatography (eluent: petroleum ether/ethyl acetate = 10/1) afforded the product **1q** as a liquid; yield: 74.9 mg (81% for 2 steps); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.44–7.25 (m, 5H, Ar-H), 4.58 (d, J = 4.8 Hz, 2H, CH_2O), 2.14–2.05 (m, 1H, OH), 1.97 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 201.7 (q, J = 4.2 Hz), 132.3, 128.8, 128.3, 126.6, 123.5 (q, J = 272.3 Hz), 112.6, 98.7 (q, J = 34.4 Hz), 61.2, 13.1; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −65.3; IR (neat): ν = 3346, 2933, 1969, 1600, 1497, 1463, 1379, 1298, 1169, 1119, 1090, 1034 cm^{-1}; MS (EI, 70 eV): m/z (%) = 228 (M^+, 39.3), 178 (100); HR-MS: m/z = 228.0763, calcd. for $C_{12}H_{11}F_3O$ (M^+): 228.0762.

Methyl 2-Butyl-4-phenyl-5-hydroxypenta-2,3-dienoate (5a)



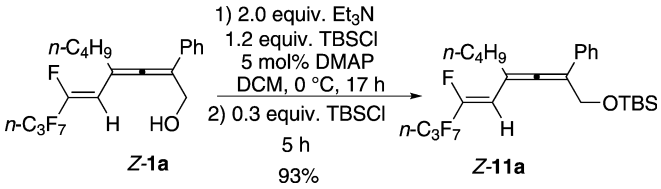
Compound **5a** was prepared according to the literature procedure.[21] To a flask was added **5aa** (401.1 mg, 2 mmol), MeOH (10 mL), and $Pd(PPh_3)_4$ (46.6 mg, 0.04 mmol) sequentially. The resulting mixture was degassed with CO for three times, and then stirred at room temperature under

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200 psi of CO in an autoclave for 5.5 h. After removal of the solvent under vacuum, the residue was purified by flash chromatography on silica gel (petroleum ether/diethyl ether = 5/1 to 4/1) to afford **5a** as a liquid; yield: 0.5211 g (100%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.45–7.23 (m, 5H, Ar-H), 4.63 (s, 2H, CH_2O), 3.74 (s, 3H, CH_3), 2.44–2.32 (m, 2H, CH_2), 2.14 (bs, 1H, OH), 1.52–1.28 (m, 4H, $CH_2\times2$), 0.88 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 210.1, 167.3, 132.8, 128.7, 127.8, 126.4, 110.6, 104.9, 61.5, 52.3, 30.2, 28.7, 22.3, 13.8; IR (neat): ν = 3417, 2955, 2919, 2872, 1944, 1716, 1496, 1453, 1436, 1267, 1133, 1087 cm^{-1}; MS (EI, 70 eV): m/z (%) = 260 (M^+, 2.2), 141 (100); HR-MS: m/z = 260.1413, calcd. for $C_{16}H_{20}O_3$ (M^+): 260.1412.

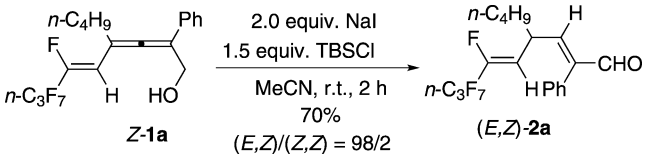
4-(*n*-Butyl)-6,7,7,8,8,9,9,9-octafluoro-2-phenylnona-2,3,5(*Z*)-trienyl *tert*-Butyl(dimethylsilyl) Ether (*Z*-11a)



To a dry Schlenk tube were added *Z*-**1a** (415.6 mg, 1.0 mmol), DCM (10 mL), Et_3N (0.28 mL, d = 0.726 g mL^{-1}, 203.3 mg, 2 mmol), TBSCl (188.1 mg, 1.2 mmol), and DMAP (6.2 mg, 0.05 mmol) sequentially at 0 °C under a nitrogen atmosphere. After 17 h, the reaction mixture was treated with TBSCl (45.5 mg, 0.3 mmol) and stirred for 5 h. After quenching with water (5 mL), the resulting mixture was extracted with diethyl ether (20 mL × 3) and dried over anhydrous Na_2SO_4. Filtration, evaporation and column chromatography on silica gel (eluent: petroleum ether) afforded *Z*-**11a** as an oil; yield: 492.7 mg (93%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.44–7.29 (m, 4H, Ar-H), 7.28–7.20 (m, 1H, Ar-H), 5.92 (d, J = 35.7 Hz, 1H, CH=), 4.60 (s, 2H, CH_2O), 2.38 (t, J = 7.7 Hz, 2H, CH_2), 1.58–1.30 (m, 4H, $CH_2\times2$), 0.94–0.80 (m, 12H, $CH_3\times4$), 0.07 (s, 6H, $CH_3\times2$); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 208.7–208.5 (m), 134.1, 128.5, 128.4 (d, J = 1.3 Hz), 126.6, 112.5–112.2 (m), 108.2 (d, J = 2.0 Hz), 103.0 (t, J = 4.2 Hz), 62.2, 31.5 (d, J = 4.2 Hz), 30.4 (d, J = 1.3 Hz), 25.7, 22.3 (d, J = 2.0 Hz), 18.1, 13.9, 13.8, −5.4; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 (t, J = 8.7 Hz, 3F), −117.9 ~(−118.2) (m, 2F), −127.2 ~(−127.4) (m, 2F), −128.2 ~(−127.6) (m, 1F); IR (neat): ν = 2958, 2932, 2860, 1935, 1681, 1496, 1472, 1361, 1256, 1231, 1187, 1120, 1051 cm^{-1}; MS (ESI): m/z (%) = 529 (M^++H), 551 (M^++Na, 100); HR-MS: m/z = 551.1990, calcd. for $C_{25}H_{32}F_8NaOSi$ (M^+): 551.1987.

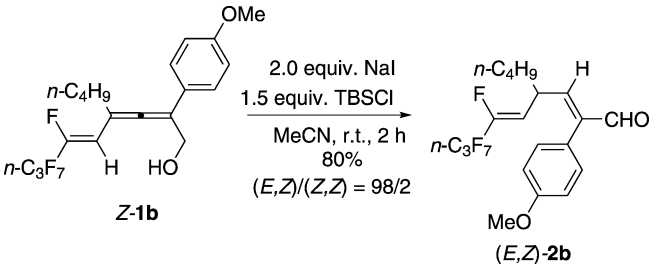
Typical Procedure 2: Synthesis of 6-Fluoro-6-perfluoroalkyl-2(*E*),5(*Z*)-dienals 2

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-phenyl-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2a]: To a dry Schlenk tube was added NaI (60.7 mg, 0.40 mmol), *Z*-**1a** (83.0 mg, 0.20 mmol), MeCN (2.0 mL), and TBSCl (46.2 mg, 0.31 mmol) at room temperature under a nitrogen atmosphere. Then the reaction mixture was stirred at room temperature for 2 h. After quenching with water (10 mL), the resulting mixture was extracted with diethyl ether (15 mL × 3), washed with brine, and dried over anhydrous Na_2SO_4. Filtration, evaporation and column chromatography on silica gel (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 40/1) afforded (*E*,*Z*)-**2a** as an oil; yield: 58.0 mg (70%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.65 (s, 1H, CHO), 7.48–7.33 (m, 3H, Ar-H), 7.16–7.05 (m, 2H, Ar-H), 6.48 (dd, J_1 = 10.4 Hz, J_2 = 0.8 Hz, 1H, CH=), 5.58 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.79–3.61 (m, 1H, CH), 1.65–1.45 (m, 2H, CH_2), 1.35–1.09 (m, 4H, $CH_2\times2$), 0.90–0.76 (m, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.0, 152.5, 146.1 (dt, J_1 = 260.0 Hz, J_2 = 29.1 Hz), 144.7, 131.9, 129.1, 128.5, 128.4, 116.4–116.0 (m), 35.1 (d, J = 2.4 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.5 ~(−128.8) (m, 1F); IR (neat): ν = 2933, 2862, 2715, 1698, 1633, 1495, 1444, 1350, 1224, 1187, 1120, 1079, 949 cm^{-1}; MS (EI, 70 eV): m/z (%) = 414 (M^+, 100); HR-MS: m/z = 414.1214, calcd. for $C_{19}H_{18}F_8O$ (M^+): 414.1230.



The following compounds were prepared according to Procedure 2.

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-(4-methoxyl-phenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2b]: The reaction of *Z*-**1b** (91.7 mg, 0.21 mmol), NaI (60.8 mg, 0.41 mmol), and TBSCl (46.1 mg, 0.31 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2b** (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 60/1 to 40/1) as an oil; yield: 73.0 mg (80%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.62 (s, 1H, CHO), 7.09–7.01 (m, 2H, Ar-H), 6.99–6.91 (m, 2H, Ar-H), 6.43 (d, J = 10.2 Hz, 1H, CH=), 5.57 (dd, J_1 = 32.9 Hz, J_2 = 10.1 Hz, 1H, CH=), 3.93–3.66 (m, 4H, CH_3O and CH), 1.62–1.46 (m, 2H, CH_2), 1.34–1.10 (m, 4H, $CH_2\times2$), 0.84 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.5, 159.6, 152.4, 145.9 (dt, J_1 = 259.7 Hz, J_2 = 28.9 Hz), 144.1, 130.3, 123.9, 116.5–116.2 (m), 113.9, 55.2, 35.1 (d, J = 2.4 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.7) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.6 ~(−129.0) (m, 1F); IR (neat): ν = 2961, 2935, 2862, 2712, 1696, 1608, 1514, 1467, 1350, 1293, 1245, 1228, 1186, 1120, 1074, 1033 cm^{-1}; MS (EI, 70 eV): m/z (%) = 444 (M^+, 100);

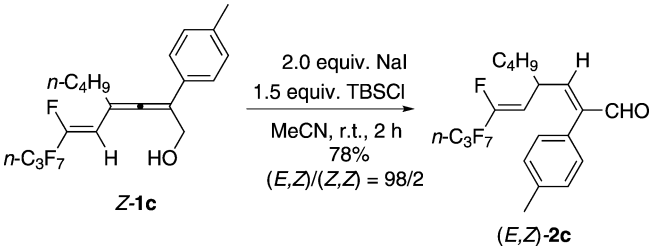


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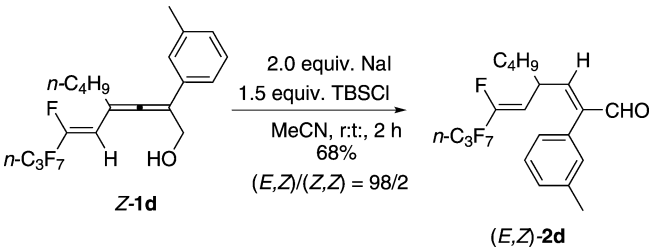
HR-MS: m/z = 444.1336, calcd. for $C_{20}H_{20}F_8O_2$ (M^+): 444.1336.

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-(4-methylphenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2c]: The reaction of *Z*-**1c** (85.7 mg, 0.20 mmol), NaI (60.8 mg, 0.41 mmol), and TBSCl



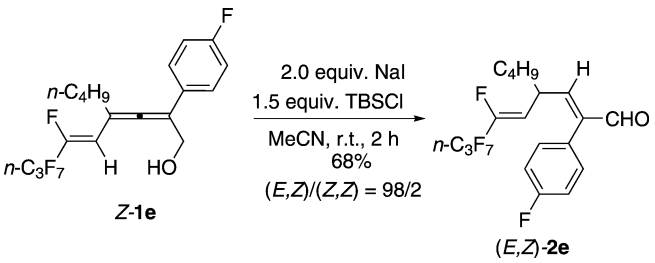
(45.8 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2c** (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 40/1) as an oil; yield: 67.0 mg (78%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.23 (d, J = 7.8 Hz, 2H, Ar-H), 6.99 (d, J = 8.1 Hz, 2H, Ar-H), 6.45 (d, J = 10.5 Hz, 1H, CH=), 5.56 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.82–3.64 (m, 1H, CH), 2.38 (s, 3H, CH_3), 1.65–1.45 (m, 2H, CH_2), 1.32–1.10 (m, 4H, $CH_2\times2$), 0.84 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.4, 152.4 (d, J = 1.4 Hz), 145.9 (dt, J_1 = 260.0 Hz, J_2 = 29.5 Hz), 144.6, 138.2, 129.2, 128.9, 128.8, 116.5–116.1 (m), 35.1 (d, J = 1.0 Hz), 34.3, 28.8, 22.2, 21.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.6 ~(−128.9) (m, 1F); IR (neat): ν = 2961, 2931, 2862, 2711, 1698, 1636, 1515, 1467, 1350, 1311, 1221, 1187, 1156, 1120, 1073, 1032 cm^{-1}; MS (EI, 70 eV): m/z (%) = 428 (M^+, 100); HR-MS: m/z = 428.1401, calcd. for $C_{20}H_{20}F_8O$ (M^+): 428.1386.

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-(3-methylphenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2d]: The reaction of *Z*-**1d** (85.8 mg, 0.20 mmol), NaI (61.0 mg, 0.41 mmol), and TBSCl



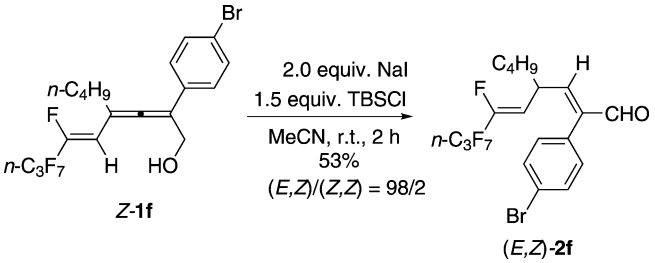
(45.0 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2d** (eluent: 30–60 °C petroleum ether/ethyl acetate = 100/1 to 60/1) as an oil; yield: 58.5 mg (68%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.35–7.27 (m, 1H, Ar-H), 7.23–7.14 (m, 1H, Ar-H), 6.95–6.85 (m, 2H, Ar-H), 6.46 (d, J = 9.9 Hz, 1H, CH=), 5.56 (dd, J_1 = 32.7 Hz, J_2 = 10.2 Hz, 1H, CH=), 3.80–3.62 (m, 1H, CH), 2.36 (s, 3H, CH_3), 1.70–1.46 (m, 2H, CH_2), 1.35–1.10 (m, 4H, $CH_2\times2$), 0.84 (t, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.2, 152.5, 145.9 (dt, J_1 = 260.1 Hz, J_2 = 29.3 Hz), 144.8, 138.1, 131.8, 129.7, 129.1, 128.3, 126.1, 116.5–116.1 (m), 35.1 (d, J = 2.0 Hz), 34.3, 28.8, 22.1, 21.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.8 ~(−129.1) (m, 1F); IR (neat): ν = 2961, 2933, 2862, 1698, 1467, 1350, 1230, 1186, 1120, 1080, 1032 cm^{-1}; MS (EI, 70 eV): m/z (%) = 428 (M^+, 27.48), 105 (100); HR-MS: m/z = 428.1386, calcd. for $C_{20}H_{20}F_8O$ (M^+): 428.1386.

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-(4-fluoro-phenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2e]: The reaction of *Z*-**1e** (87.2 mg, 0.20 mmol), NaI (60.8 mg, 0.41 mmol), and TBSCl



(46.1 mg, 0.31 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2e** (eluent: 30–60 °C petroleum ether/ethyl acetate = 100/1 to 80/1) as an oil; yield: 58.9 mg (68%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.18–7.03 (m, 4H, Ar-H), 6.49 (d, J = 10.2 Hz, 1H, CH=), 5.57 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.78–3.60 (m, 1H, CH), 1.70–1.48 (m, 2H, CH_2), 1.32–1.10 (m, 4H, $CH_2\times2$), 0.84 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 192.9, 164.4, 161.1, 153.1, 146.2 (dt, J_1 = 259.9 Hz, J_2 = 29.3 Hz), 143.8, 130.9 (d, J = 8.1 Hz), 127.7 (d, J = 3.5 Hz), 116.2–115.9 (m), 115.6 (d, J = 21.2 Hz), 35.1 (d, J = 1.7 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −113.1 ~(−113.2) (m, 1F), −118.6 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.4 ~(−128.75) (m, 1F); IR (neat): ν = 2962, 2934, 2863, 2715, 1697, 1604, 1512, 1350, 1225, 1187, 1159, 1120, 1071 cm^{-1}; MS (EI, 70 eV): m/z (%) = 432 (M^+, 84.99), 178 (100); HR-MS: m/z = 432.1122, calcd. for $C_{19}H_{17}F_9O$ (M^+): 432.1136.

4-Butyl-6c,7,7,8,8,9,9,9-octafluoro-2-(4-bromophenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2f]: The reaction of *Z*-**1f** (91.8 mg, 0.19 mmol) and NaI (56.7 mg, 0.37 mmol) and



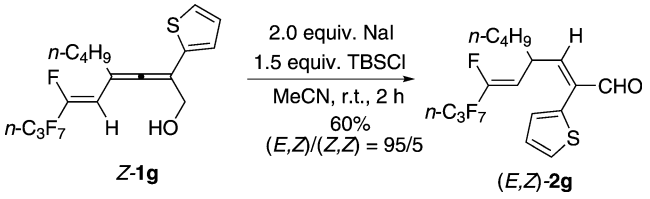
TBSCl (42.0 mg, 0.28 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2f** (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 60/1) as an oil; yield: 48.4 mg (53%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.62 (s, 1H, CHO), 7.60–7.53 (m, 2H, Ar-H), 7.03–6.94 (m, 2H, Ar-H), 6.50 (dd, J_1 = 10.4 Hz, J_2 = 0.5 Hz, 1H, CH=

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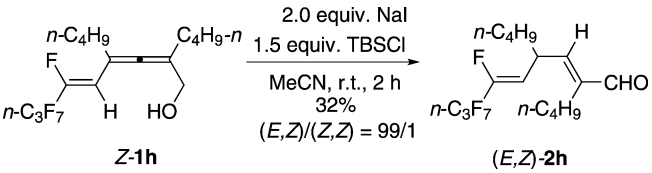
), 5.56 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.75–3.57 (m, 1H, CH), 1.60–1.47 (m, 2H, CH_2), 1.32–1.11 (m, 4H, $CH_2\times2$), 0.85 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 192.6, 153.1, 143.6, 131.8, 130.8, 130.7, 122.8, 116.1–115.8 (m), 35.2 (d, J = 1.4 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.6 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.1 ~(−128.4) (m, 1F); IR (neat): ν = 2955, 2925, 2854, 2712, 1699, 1638, 1590, 1489, 1465, 1349, 1220, 1187, 1156, 1121, 1066, 1012 cm^{-1}; MS (EI, 70 eV): m/z (%) = 495 (M^+-(^{81}Br)+1, 9.74), 494 (M^+(^{81}Br), 45.03), 493 (M^+(^{79}Br)+1, 10.25), 492 (M^+(^{79}Br), 46.62), 57 (100); HR-MS: m/z = 492.0334, calcd. for $C_{19}H_{17}{}^{79}BrF_8O$ (M^+(^{79}Br)): 492.0335.

4-Butyl-6,7,7,8,8,9,9,9-octafluoro-2-thienyl-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2g]: The reaction of *Z*-**1g** (84.1 mg, 0.20 mmol), NaI (60.8 mg, 0.41 mmol), and TBSCl (45.6 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2g** (eluent: 30–60 °C petroleum ether/ethyl acetate = 150/1) as an oil; yield: 50.1 mg (60%); (*E*,*Z*)/(*Z*,*Z*) = 95/5 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.60 (s, 1H, CHO), 7.47 (dd, J_1 = 3.6 Hz, J_2 = 2.4 Hz, 1H, Ar-H), 7.15–7.08 (m, 2H, Ar-H), 6.47 (d, J = 10.2 Hz, 1H, CH=), 5.60 (dd, J_1 = 32.9 Hz, J_2 = 9.8 Hz, 1H, CH=), 4.16–4.00 (m, 1H, CH), 1.70–1.53 (m, 2H, CH_2), 1.38–1.15 (m, 4H, $CH_2\times2$), 0.87 (t, J = 6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 192.3, 153.0, 136.9, 131.2, 128.6, 127.5, 127.0, 116.2–115.8 (m), 35.2 (d, J = 3.9 Hz), 34.5, 28.8, 22.3, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.7 ~(−80.9) (m, 3F), −118.5 ~(−118.7) (m, 2F), −127.3 ~(−127.5) (m, 2F), −127.8 ~(−128.1) (m, 1F); IR (neat): ν = 2961, 2933, 2862, 2718, 1700, 1440, 1351, 1228, 1187, 1155, 1122, 1083, 1029 cm^{-1}; MS (EI, 70 eV): m/z (%) = 420 (M^+, 76.52), 109 (100); HR-MS: m/z = 420.0786, calcd. for $C_{17}H_{16}F_8OS$ (M^+): 420.0794.



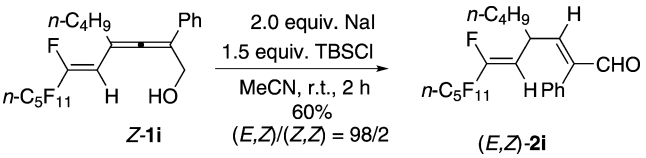
Z-**1g** (*E*,*Z*)-**2g**

2,4-Dibutyl-6,7,7,8,8,9,9,9-octafluoro-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2h]: The reaction of *Z*-**1h** (78.0 mg, 0.20 mmol), NaI (60.4 mg, 0.40 mmol), and TBSCl (45.1 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2h** (eluent: 30–60 °C petroleum ether/ethyl ether = 100/1 to petroleum ether/ethyl acetate = 60/1) as an oil; yield: 25.1 mg (32%); (*E*,*Z*)/(*Z*,*Z*) = 99/1 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.39 (s, 1H, CHO), 6.19 (d, J = 10.2 Hz, 1H, CH=), 5.52 (dd, J_1 = 32.4 Hz, J_2 = 10.2 Hz, 1H, CH=), 3.89–3.73 (m, 1H, CH), 2.38–2.16 (m, 2H, CH_2), 1.68–1.50 (m, 2H, CH_2), 1.42–1.18 (m, 8H, $CH_2\times4$), 0.97–0.84 (m, 6H, $CH_3\times2$); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 194.8, 151.8–151.7 (m), 144.5, 116.7–116.3 (m), 34.4–34.2 (m), 30.1, 29.0, 24.1, 22.8, 22.3, 13.8, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.6 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −129.7 ~(−130.0) (m, 1F); IR (neat): ν = 2961, 2934, 2864, 1694, 1642, 1468, 1351, 1229, 1187, 1155, 1122 cm^{-1}; MS (EI, 70 eV): m/z (%) = 394 (M^+, 12.83), 111 (100); HR-MS: m/z = calcd. for $C_{17}H_{22}F_8O$ (M^+): 394.1543.



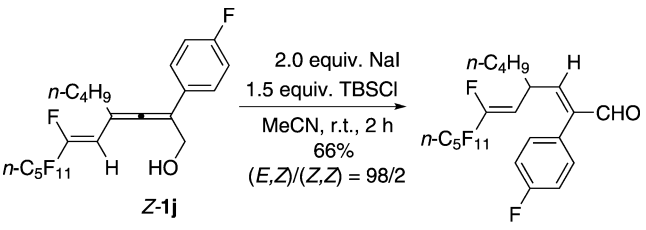
Z-**1h** (*E*,*Z*)-**2h**

4-Butyl-6,7,7,8,8,9,9,9,10,10,11,11,11-dodecafluoro-2-phenyl-2(*E*),5(*Z*)-undecadienal [(*E*,*Z*)-2i]: The reaction of *Z*-**1i** (102.8 mg, 0.20 mmol), NaI (60.9 mg, 0.40 mmol), and TBSCl (45.7 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2i** (eluent: 30–60 °C petroleum ether/ethyl acetate = 100/1 to 80/1) as an oil; yield: 62.1 mg (60%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.65 (s, 1H, CHO), 7.48–7.34 (m, 3H, Ar-H), 7.14–7.06 (m, 2H, Ar-H), 6.49 (d, J = 10.2 Hz, 1H, CH=), 5.57 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.78–3.63 (m, 1H, CH), 1.64–1.45 (m, 2H, CH_2), 1.30–1.08 (m, 4H, $CH_2\times2$), 0.83 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.2, 152.7, 146.2 (dt, J_1 = 259.4 Hz, J_2 = 29.0 Hz), 144.7, 131.8, 129.0, 128.44, 128.37, 116.4–116.1 (m), 35.1 (d, J = 1.7 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−81.0) (m, 3F), −117.5 ~(−117.8) (m, 2F), −122.9 ~(−123.2) (m, 2F), −123.2 ~(−123.4) (m, 2F), −126.2 ~(−126.5) (m, 2F), −128.3 ~(−128.7) (m, 1F); IR (neat): ν = 2962, 2934, 2863, 2710, 1698, 1636, 1601, 1497, 1468, 1445, 1353, 1317, 1239, 1203, 1143, 1108, 1081, 1028 cm^{-1}; MS (EI, 70 eV): m/z (%) = 514 (M^+, 71.56), 160 (100); HR-MS: m/z = 514.1163, calcd. for $C_{21}H_{18}F_{12}O$ (M^+): 514.1166.



Z-**1i** (*E*,*Z*)-**2i**

4-Butyl-6,7,7,8,8,9,9,10,10,11,11,11-dodecafluoro-2-(4-fluoro-phenyl)-2(*E*),5(*Z*)- undecadienal [(*E*,*Z*)-2j]: The reaction of *Z*-**1j** (106.8 mg, 0.20 mmol), NaI (61.0 mg, 0.41 mmol), and TBSCl (45.2 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2j** (eluent: 30–60 °C petroleum ether/ethyl acetate = 100/1 to 60/1 to 40/1) as an oil; yield: 70.3 mg (66%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.17–7.04 (m, 4H, Ar-H), 6.49 (d, J = 10.2 Hz, 1H, CH=), 5.57 (dd, J_1 = 32.7 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.76–3.61 (m, 1H, CH), 1.65–1.48 (m, 2H, CH_2), 1.36–1.09 (m, 4H, $CH_2\times2$), 0.84 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$):



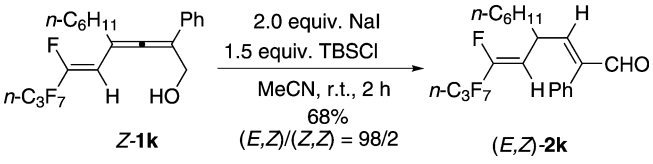
Z-**1j** (*E*,*Z*)-**2j**

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16154169, 2014, 6, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300835 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

δ = 193.0, 162.7 (d, J = 246.3 Hz), 153.2, 146.3 (dt, J_1 = 259.2 Hz, J_2 = 28.3 Hz), 143.7, 130.9 (d, J = 8.0 Hz), 127.6 (d, J = 3.5 Hz), 116.3–115.9 (m), 115.7 (d, J = 21.6 Hz), 35.1 (d, J = 2.0 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −113.1 ~(−113.2) (m, 1F), −117.6 ~(−117.8) (m, 2F), −122.9 ~(−123.4) (m, 4F), −126.2 ~(−126.5) (m, 2F), −128.2 ~(−128.5) (m, 1F); IR (neat): ν = 2963, 2934, 2863, 1698, 1604, 1512, 1456, 1353, 1317, 1238, 1200, 1155, 1143, 1108, 1074 cm^{-1}; MS (EI, 70 eV): m/z (%) = 532 (M^+, 100); HR-MS: m/z = 532.1073, calcd. for $C_{21}H_{17}F_{13}O$ (M^+): 532.1072.

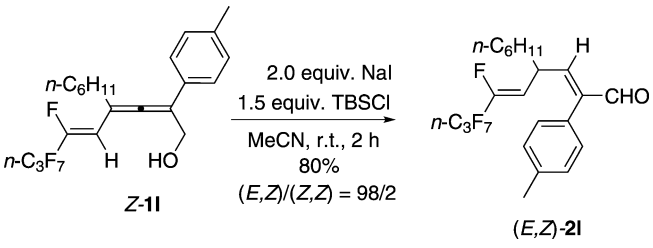
4-Hexyl-6,7,7,8,8,9,9,9-octafluoro-2-phenyl-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2k]: The reaction of *Z*-**1k** (89.0 mg, 0.20 mmol), NaI (60.7 mg, 0.41 mmol), and TBSCl (45.8 mg,



Z-**1k** (*E*,*Z*)-**2k**

0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2k** (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 60/1) as an oil; yield: 60.2 mg (68%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.65 (s, 1H, CHO), 7.47–7.34 (m, 3H, Ar-H), 7.15–7.06 (m, 2H, Ar-H), 6.48 (d, J = 10.2 Hz, 1H, CH=), 5.56 (dd, J_1 = 33.0 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.79–3.62 (m, 1H, CH), 1.59–1.44 (m, 2H, CH_2), 1.32–1.08 (m, 8H, $CH_2\times4$), 0.85 (t, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$, a signal at 29 ppm belongs to residual petroleum ether): δ = 193.2, 152.7, 144.6, 131.8, 129.0, 128.45, 128.38, 116.4–116.0 (m), 35.1 (d, J = 1.1 Hz), 34.5, 31.4, 28.7, 26.6, 22.5, 14.0; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.6 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.5 ~(−128.8) (m, 1F); IR (neat): ν = 3060, 2930, 2859, 2710, 1694, 1635, 1601, 1497, 1467, 1445, 1350, 1231, 1156, 1120, 1078, 1028 cm^{-1}; MS (EI, 70 eV): m/z (%) = 442 (M^+, 100); HR-MS: m/z = 442.1541, calcd. for $C_{21}H_{22}F_8O$ (M^+): 442.1543.

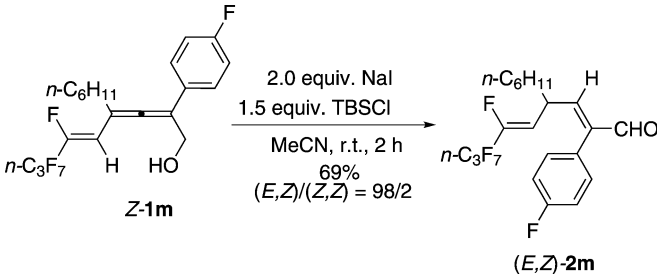
4-Hexyl-6,7,7,8,8,9,9,9-octafluoro-2-(4-methylphenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2l]: The reaction of *Z*-**1l** (90.8 mg, 0.20 mmol), NaI (60.8 mg, 0.41 mmol), and TBSCl



Z-**1l** (*E*,*Z*)-**2l**

(46.1 mg, 0.31 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2l** (eluent: 30–60 °C petroleum ether/ethyl acetate = 80/1 to 60/1) as an oil; yield: 72.8 mg (80%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.23 (d, J = 7.8 Hz, 2H, Ar-H), 7.00 (d, J = 8.1 Hz, 2H, Ar-H), 6.45 (d, J = 9.9 Hz, 1H, CH), 5.57 (dd, J_1 = 32.7 Hz, J_2 = 10.2 Hz, 1H, CH=), 3.82–3.65 (m, 1H, CH), 2.37 (s, 3H, CH_3), 1.66–1.45 (m, 2H, CH_2), 1.32–1.10 (m, 8H, $CH_2\times4$), 0.85 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.4, 152.5, 146.0 (dt, J_1 = 260.2 Hz, J_2 = 28.9 Hz), 144.5, 138.2, 129.1, 128.9, 128.8, 116.6–116.2 (m), 35.1 (d, J = 1.5 Hz), 34.6, 31.4, 28.7, 26.6, 22.4, 21.2, 13.9; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.9 ~(−81.0) (m, 3F), −118.6 ~(−118.8) (m, 2F), −127.5 ~(−127.6) (m, 2F), −128.6 ~(−129.0) (m, 1F); IR (neat): ν = 2961, 2930, 2859, 2710, 1696, 1634, 1515, 1460, 1351, 1309, 1228, 1187, 1120, 1074, 1032 cm^{-1}; MS (EI, 70 eV): m/z (%) = 456 (M^+, 12.94), 43 (100); HR-MS: m/z = 456.1698, calcd. for $C_{22}H_{24}F_8O$ (M^+): 456.1699.

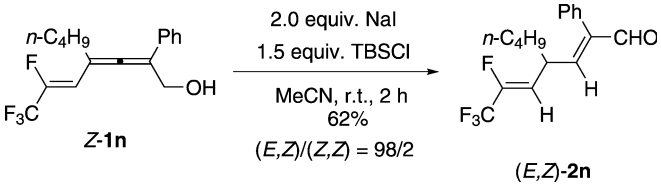
4-Hexyl-6,7,7,8,8,9,9,9-octafluoro-2-(4-fluorophenyl)-2(*E*),5(*Z*)-nonadienal [(*E*,*Z*)-2m]: The reaction of *Z*-**1m** (93.1 mg, 0.20 mmol), NaI (59.9 mg, 0.40 mmol), and TBSCl



Z-**1m** (*E*,*Z*)-**2m**

(45.7 mg, 0.30 mmol) in MeCN (2.0 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2m** (eluent: 30–60 °C petroleum ether/ethyl acetate = 100/1 to 80/1 to 60/1) as an oil; yield: 64.2 mg (69%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.17–7.04 (m, 4H, Ar-H), 6.49 (d, J = 10.2 Hz, 1H, CH=), 5.57 (dd, J_1 = 33.0 Hz, J_2 = 9.9 Hz, 1H, CH=), 3.77–3.60 (m, 1H, CH), 1.63–1.47 (m, 2H, CH_2), 1.32–1.11 (m, 8H, $CH_2\times4$), 0.85 (t, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.0, 162.7 (d, J = 246.5 Hz), 153.2, 146.1 (dt, J_1 = 259.1 Hz, J_2 = 28.6 Hz), 143.7, 130.9 (d, J = 7.5 Hz), 127.6 (d, J = 3.4 Hz), 116.2–115.9 (m), 115.6 (d, J = 21.2 Hz), 35.1 (d, J = 1.7 Hz), 34.5, 31.4, 28.7, 26.6, 22.5, 13.9; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −113.1 ~(−113.3) (m, 1F), −118.6 ~(−118.8) (m, 2F), −127.4 ~(−127.6) (m, 2F), −128.4 ~(−128.8) (m, 1F); IR (neat): ν = 2932, 2860, 2712, 1697, 1604, 1512, 1468, 1351, 1228, 1187, 1159, 1121, 1096, 1072 cm^{-1}; MS (EI, 70 eV): m/z (%) = 460 (M^+, 100); HR-MS: m/z = 460.1428, calcd. for $C_{21}H_{21}F_9O$ (M^+): 460.1449.

4-Butyl-6,7,7,7-tetrafluoro-2-phenyl-2(*E*),5(*Z*)-heptadienal [(*E*,*Z*)-2n]: The reaction of *Z*-**1n** (314.0 mg, 1.0 mmol), NaI (303.1 mg, 2.0 mmol), and TBSCl (228.1 mg, 1.5 mmol) in MeCN (10 mL) at room temperature for 2 h afforded (*E*,*Z*)-



Z-**1n** (*E*,*Z*)-**2n**

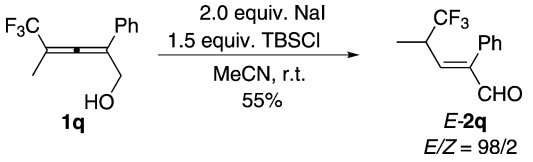
2n (eluent: 30–60 °C petroleum ether/ethyl ether = 100/1 to 60/1 to petroleum ether/ethyl acetate = 30/1) as an oil; yield: 194.1 mg (62%); (*E*,*Z*)/(*Z*,*Z*) = 98/2 determined by ^{1}H NMR analysis of the crude product; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.63 (s, 1H, CHO), 7.48–7.31 (m, 3H, Ar-H), 7.18–7.06

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(m, 2H, Ar-H), 6.47 (d, J=9.9 Hz, 1H, CH=), 5.51 (dd, J_1=33.3 Hz, J_2=9.9 Hz, 1H, CH=), 3.74–3.57 (m, 1H, CH), 1.62–1.45 (m, 2H, CH_2), 1.30–1.06 (m, 4H, $CH_2\times2$), 0.83 (t, J=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=193.1, 152.9, 146.0 (dq, J_1=258.9 Hz, J_2=38.4 Hz), 144.5, 131.9, 129.0, 128.4, 128.3, 118.2 (qd, J_1=269.7 Hz, J_2=42.0 Hz), 113.4–113.1 (m), 34.7 (d, J=1.7 Hz), 34.2, 28.8, 22.2, 13.6; ^{19}F NMR (282 MHz, $CDCl_3$): δ=−72.9 (d, J=10.7 Hz, 3F), −133.6 (dq, J_1=33.3 Hz, J_2=11.6 Hz, 1F); IR (neat): ν=2960, 2933, 2861, 2709, 1696, 1636, 1497, 1467, 1444, 1362, 1342, 1199, 1145, 1080 cm^{-1}; MS (EI, 70 eV): m/z (%)=314 (M^+, 31.64), 105 (100); HR-MS: m/z=314.1297, calcd. for $C_{17}H_{18}F_4O$ (M^+): 314.1294.

4-Methyl-2-phenyl-5,5,5-trifluoro-2(*E*)-pentenal [(*E*)-2q]: The reaction of **1q** (411.1 mg, 1.8 mmol), NaI (541.8 mg,

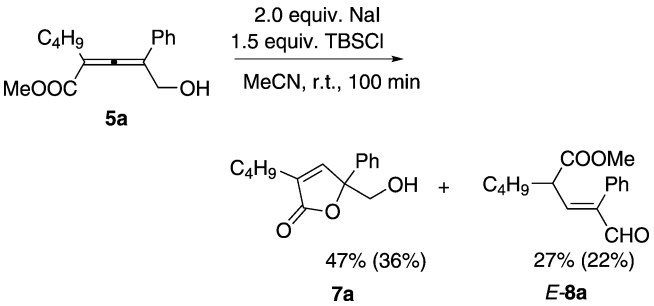


3.6 mmol), and TBSCl (410.4 mg, 2.7 mmol) in MeCN (15 mL) at room temperature for 2 h afforded *E*-**2q** as a liquid (eluent: 30–60 °C petroleum ether/ethyl acetate=100/1 to 60/1); yield: 226.4 mg (55%) (there is an unidentified product but is not the stereoisomer in the crude product); ^{1}H NMR (300 MHz, $CDCl_3$): δ=9.67 (s, 1H, CHO), 7.54–7.27 (m, 3H, Ar-H), 7.24–7.04 (m, 2H, Ar-H), 6.57 (d, J=10.5 Hz, 1H, CH=), 3.44–3.22 (m, 1H, CH), 1.25 (d, J=6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=192.7, 146.7, 146.5–146.2 (m), 131.4, 128.9, 128.6, 126.4 (q, J=277.9 Hz), 38.8 (q, J=28.3 Hz), 13.8–13.5 (m); ^{19}F NMR (282 MHz, $CDCl_3$): δ=−71.4 (d, J=8.5 Hz); IR (neat): ν=2713, 1701, 1497, 1461, 1445, 1386, 1369, 1350, 1259, 1226, 1176, 1132, 1086, 1055, 1019 cm^{-1}; MS (EI, 70 eV): m/z (%)=228 (M^+, 100); HR-MS: m/z=228.0762, calcd. for $C_{12}H_{11}F_3O$ (M^+): 228.0762.

Control Experiments

3-Butyl-5-(hydroxymethyl)-5-phenyl-2(5*H*)-furanone (7a) and methyl 2-butyl-5-oxo-4-phenyl-3(*Z*)-pentenoate [(*E*)-8a]:

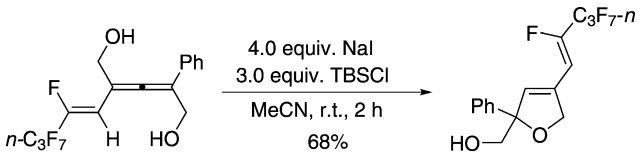
The reaction of **5a** (51.5 mg, 0.20 mmol), NaI (60.1 mg, 0.4 mmol), and TBSCl (46.1 mg, 0.3 mmol) in MeCN (2.0 mL) at room temperature for 100 min afforded (eluent: petroleum ether/ethyl acetate=40/1 to 12/1 to 10/1 to 3/1) **7a**; yield: 17.4 mg (36%) and (*E*)-**8a**;yield: 11.5 mg (22%) [yields determined by ^{1}H NMR analysis of crude product: **7a** (47%)+**8a** (27%)+**9a** (6%)].



7a: oil; ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.45–7.31 (m, 5H, Ar-H), 7.27 (t, J=1.4 Hz, 1H, CH=), 4.10–3.83 (m, 2H, CH_2O), 2.62 (brs, 1H, OH), 2.41–2.21 (m, 2H, CH_2), 1.64–1.48 (m, 2H, CH_2), 1.45–1.28 (m, 2H, CH_2), 0.91 (t, J=7.4 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=173.2, 148.2, 136.7, 135.0, 128.8, 128.5, 125.3, 89.7, 67.3, 29.4, 25.0, 22.2, 13.7; IR (neat): ν=3419, 2967, 2930, 2871, 1754, 1493, 1449, 1077, 1037 cm^{-1}; MS (EI, 70 eV): m/z (%)=215 ([M−CH_2OH]$^+$, 100); MS (ESI): m/z (%)=247 (M^++H, 100); HR-MS: m/z=247.1332, calcd. for $C_{15}H_{19}O_3$ (M^++H): 247.1329.

(*E*)-**8a:** oil; ^{1}H NMR (300 MHz, $CDCl_3$): δ=9.68 (s, 1H, CHO), 7.47–7.33 (m, 3H, Ar-H), 7.18–7.11 (m, 2H, Ar-H), 6.71 (d, J=10.5 Hz, 1H, CH=), 3.74 (s, 3H, CH_3O), 3.55–3.43 (m, 1H, CH), 1.92–1.75 (m, 1H, CH_2), 1.71–1.54 (m, 1H, CH_2), 1.31–1.07 (m, 4H, $CH_2\times2$), 0.83 (t, J=6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=193.2, 172.9, 151.3, 145.0, 131.8, 129.3, 128.4, 128.3, 52.3, 45.6, 32.4, 28.9, 22.2, 13.7; IR (neat): ν=2956, 2860, 1737, 1696, 1636, 1497, 1444, 1197, 1168, 1080 cm^{-1}; MS (EI, 70 eV): m/z (%) 260 (M^+, 21.4), 115 (100); HR-MS: m/z=260.1414, calcd. for $C_{16}H_{20}O_3$ (M^+): 260.1412.

2-Hydroxymethyl-2-phenyl-4-[2,3,3,4,4,5,5,5-octafluoro-1(*Z*)-pentenyl]-2,5-dihydrofuran [(*Z*)-10a]: The reaction of *Z*-**1o** (140.5 mg, 0.36 mmol), NaI (210.8 mg, 1.4 mmol), and

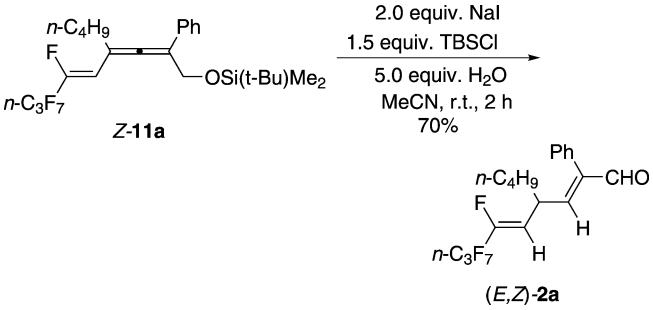


TBSCl (162.5 mg, 1.08 mmol) in MeCN (4 mL) at room temperature for 2 h afforded *Z*-**10a** (eluent: petroleum ether/ethyl acetate=10/1 to 5/1 for first round; DCM to DCM/ethyl acetate=5/1 for second round)as an oil; yield: 95.1 mg (68%); ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.42–7.25 (m, 5H, Ar-H), 6.36 (s, 1H, CH=), 6.26 (d, J=36.9 Hz, 1H, CH=), 5.08 (d, J=12.6 Hz, 1H, CH_2), 4.97 (d, J=12.9 Hz, 1H, CH_2), 3.91–3.73 (m, 2H, CH_2), 2.41 (dd, J_1=8.7 Hz, J_2=4.5 Hz, 1H, OH); ^{13}C NMR (75 MHz, $CDCl_3$): δ=141.0, 134.9 (d, J=4.8 Hz), 131.7 (d, J=4.1 Hz), 128.6, 127.7, 124.9, 107.5–107.1 (m), 94.2, 75.0 (d, J=7.7 Hz), 68.5; ^{19}F NMR (282 MHz, $CDCl_3$): δ=−80.8 ~(−80.9) (m, 3F), −118.4 ~(−118.6) (m, 2F), −125.5 ~(−125.8) (m, 1F), −127.2 ~(−127.3) (m, 2F); IR (neat): ν=3426, 3072, 2921, 2868, 1687, 1491, 1449, 1364, 1230, 1155, 1120, 1043 cm^{-1}; MS (EI, 70 eV): m/z (%)=357 ([M−CH_2OH]$^+$, 100); elemental analysis calcd. for $C_{16}H_{12}F_8O_2$ (MW=388): C 49.50, H 3.12; found: C 49.49, H 3.25.

Reaction of (*Z*)-11a to Afford (*E*,*Z*)-2a: The reaction of *Z*-**11a** (79.8 mg, 0.15 mmol), NaI (45.8 mg, 0.3 mmol), H_2O (13.5 μL, 0.75 mmol), and TBSCl (34.1 mg, 0.225 mmol) in MeCN (1.5 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2a** (eluent: petroleum ether/ethyl ether=60/1) as an oil; yield: 43.7 mg (70%); ^{1}H NMR (300 MHz, $CDCl_3$): δ=9.64 (s, 1H, CHO), 7.48–7.32 (m, 3H, Ar-H), 7.16–7.04 (m, 2H, Ar-H), 6.48 (d, J=9.9 Hz, 1H, CH=), 5.58 (dd, J_1=32.9 Hz, J_2=10.1 Hz, 1H, CH=), 3.80–3.62 (m, 1H, CH), 1.64–1.45 (m, 2H, CH_2), 1.35–1.03 (m, 4H, $CH_2\times2$), 0.94–

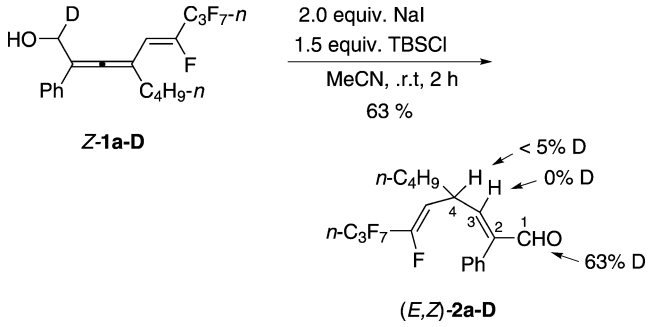
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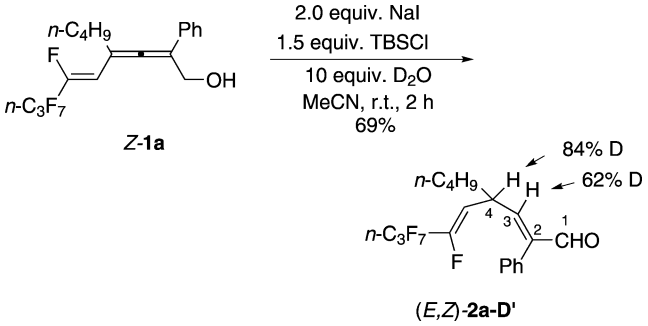
0.75 (m, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.1, 152.6, 144.7, 131.9, 129.0, 128.44, 128.37, 116.4–116.0 (m), 35.1 (d, J = 2.4 Hz), 34.3, 28.8, 22.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.8) (m, 2F), −127.3 ~(−127.6) (m, 2F), −128.5 ~(−128.8) (m, 1F); IR (neat): ν = 2961, 2933, 2862, 2710, 1698, 1497, 1361, 1221, 1187, 1120, 1079 cm^{-1}; MS (EI, 70 eV): m/z (%) = 414 (M^+, 100); HR-MS: m/z = 414.1236, calcd. for $C_{19}H_{18}F_8O$: 414.1230.

Reaction of (*Z*)-1a-D to Afford (*E*,*Z*)-2a-D: The reaction of *Z*-**1a-D** (83.8 mg, 0.2 mmol), NaI (60.8 mg, 0.4 mmol), and TBSCl (45.1 mg, 0.3 mmol) in MeCN (2 mL) at room



temperature for 2 h afforded (*E*,*Z*)-**2a-D** (eluent: petroleum ether to petroleum ether/ethyl ether = 60/1) as an oil; yield: 52.8 mg (63%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.47–7.33 (m, 3H, Ar-H), 7.15–7.06 (m, 2H, Ar-H), 6.48 (d, J = 10.2 Hz, 1H, CH=), 5.56 (dd, J_1 = 32.9 Hz, J_2 = 9.8 Hz, 1H, CH=), 3.79–3.62 (m, 0.97H, CH), 1.65–1.45 (m, 2H, CH_2), 1.35–1.10 (m, 4H, CH_2×2), 0.83 (t, J = 6.9 Hz, 3H, CH_3); ^{1}H NMR [300 MHz, $CDCl_3$, the following signal is discernible for (*E*,*Z*)-**2a**]: δ = 9.64 (s, 0.37H, CHO); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 193.0, 152.5, 146.1 (dt, J_1 = 259.4 Hz, J_2 = 29.1 Hz), 144.7, 131.9, 129.1, 128.5, 128.4, 116.4–116.0 (m), 35.1 (d, J = 1.7 Hz), 34.3, 28.8, 22.2, 13.7; ^{13}C NMR [75 MHz, $CDCl_3$, the following signal is discernible for (*E*,*Z*)-**2a-D**]: δ = 192.7 (t, J = 26.5 Hz); ^{19}F NMR (282 MHz, $CDCl_3$): δ = −80.8 ~(−80.9) (m, 3F), −118.5 ~(−118.8) (m, 2F), −127.4 ~(−127.5) (m, 2F), −128.5 ~(−128.8) (m, 1F); IR (neat): ν = 2964, 2932, 2860, 1691, 1629, 1495, 1471, 1447, 1352, 1227, 1182, 1156, 1117, 1069, 1034 cm^{-1}; MS (EI, 70 eV): m/z (%) = 415 ([M+1]$^+$, 100), 414 (M^+, 62.7); HR-MS: m/z = 415.1294, calcd. for $C_{19}H_{17}DF_8O$: 415.1293.

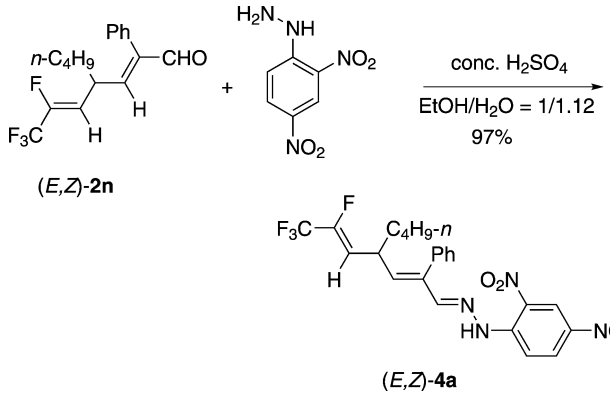
Reaction of (*Z*)-1a with 10 equiv. D_2O to Afford (*E*,*Z*)-2a-D′: The reaction of (*Z*)-**1a** (331.3 mg, 0.8 mmol), NaI (241.5 mg, 1.6 mmol), D_2O (145 μL, 8 mmol), and TBSCl (181.7 mg, 1.2 mmol) in MeCN (8 mL) at room temperature for 2 h afforded (*E*,*Z*)-**2a-D′** (eluent: 30–60 °C petroleum



ether/ethyl acetate = 80/1) as an oil; yield: 227.6 mg (69%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 9.62 (s, 1H, CHO), 7.47–7.31 (m, 3H, Ar-H), 7.18–7.06 (m, 2H, Ar-H), 5.67–5.46 (m, 1H, CH=), 1.63–1.42 (m, 2H, CH_2), 1.31–1.07 (m, 4H, CH_2×2), 0.82 (t, J = 6.8 Hz, 3H, CH_3); ^{1}H NMR [300 MHz, $CDCl_3$, the following signals are discernible for (*E*,*Z*)-**2a**] δ = 6.52–6.42 (m, 0.38H, CH=), 3.79–3.62 (m, 0.16H, CH); due to the presence of two D-incorporation, the ^{13}C NMR (75 MHz, $CDCl_3$) spectra are very complicated. These spectra are attached in the Supporting Information; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −81.0 ~(−81.1) (m, 3F), −118.6 ~(−118.9) (m, 2F), −127.5 ~(−127.7) (m, 2F), −128.8 ~(−129.2) (m, 1F); IR (neat): ν = 2961, 2933, 2862, 2715, 1698, 1496, 1361, 1230, 1187, 1121 cm^{-1}; MS (EI, 70 eV): m/z (%) = 417 ([$M(D_2)^+$+1], 20.7), 416 ([$M(D_2)$]$^+$ or [$M(D_1)^+$+1], 100), 415 ([$M(D_1)$]$^+$ or [M^++1], 66.5); HR-MS: m/z = 416.1354, calcd. for $C_{19}H_{16}D_2F_8O$: 416.1355.

Condensation Reaction of (*E*,*Z*)-4a, (*E*)-4b

4-Butyl-2-phenyl-6,7,7,7-tetrafluoro-hepta-2(*E*),5(*Z*)-dienal (2,4-Dinitrophenyl)hydrazone [(*E*,*Z*)-4a] To a dried flask equipped with an addition funnel were added (2,4-dinitrophenyl)hydrazine (91.8 mg, 0.46 mmol), concentrated H_2SO_4



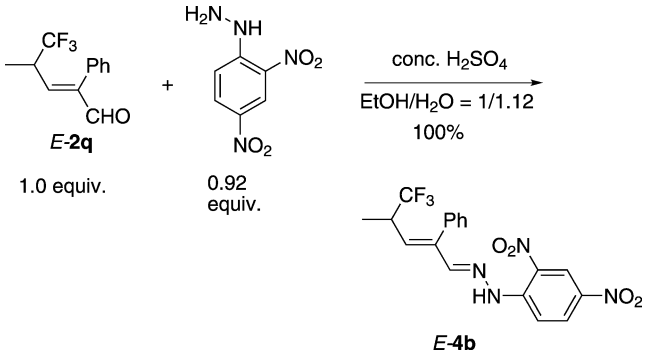
(0.75 mL), EtOH (7 mL), and H_2O (16 mL) sequentially. Then (*E*,*Z*)-**2n** (155.2 mg, 0.5 mmol) in EtOH (8 mL) was added dropwise within 8 min at room temperature. The resulting mixture was stirred at room temperature for 1.5 h, extracted with ethyl acetate (50 mL×3), washed with brine (30 mL×2), and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (eluent: petroleum ether/ethyl acetate/dichloromethane = 10/1/10) afforded (*E*,*Z*)-**4a** as an orange solid; yield: 222.9 mg (97%); mp 183.4–184.9 °C (DCM/hexane); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 11.19 (s, 1H), 9.08 (d, J = 2.4 Hz, 1H), 8.17 (dd, J_1 =

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9.5 Hz, J_2 = 2.3 Hz, 1H, Ar-H), 7.96 (s, 1H, Ar-H), 7.56–7.36 (m, 4H, Ar-H), 7.21 (d, J = 7.2 Hz, 2H, Ar-H), 5.99 (d, J = 10.5 Hz, 1H, CH=), 5.47 (dd, J_1 = 33.3 Hz, J_2 = 9.8 Hz, 1H, CH=), 3.58–3.38 (m, 1H, CH), 1.64–1.36 (m, 2H, CH_2), 1.34–1.06 (m, 4H, CH_2×2), 0.84 (t, J = 6.5 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 151.0, 145.5 (dq, J_1 = 257.9 Hz, J_2 = 38.4 Hz), 144.8, 141.4, 139.8, 138.2, 134.5, 130.0, 129.2, 128.3, 128.0, 123.2, 118.3 (qd, J_1 = 270.1 Hz, J_2 = 41.0 Hz), 114.3–113.8 (m), 34.6, 28.9, 22.2, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −72.9 (d, J = 11.8 Hz, 3F), −134.3 ~ −134.7 (m, 1F); IR (neat): ν = 3293, 1619, 1596, 1578, 1516, 1500, 1339, 1325, 1305, 1260, 1214, 1146, 1072 cm^{-1}; MS (EI, 70 eV): m/z (%) = 494 (M^+, 30.04), 311 (100); elemental analysis calcd. for $C_{23}H_{22}F_4N_4O_4$: C 55.87, H 4.48, N 11.33; found: C 55.88, H 4.70, N 11.22.

4-Methyl-2-phenyl-5,5,5-trifluoropent-2(E)-enal (2,4-dinitrophenyl)hydrazone [(*E*)-4b]: To a dried flask equipped with an addition funnel were added (2,4-dinitrophenyl)hydrazine (180.1 mg, 0.92 mmol), concentrated H_2SO_4



(1.5 mL), EtOH (20 mL), and H_2O (35 mL) sequentially. Then (*E*)-**q** (226.4 mg, 1.0 mmol) in EtOH (10 mL) was added dropwise within 10 min at room temperature. The resulting mixture was stirred at room temperature for 1.5 h, extracted with ethyl acetate (50 mL × 3), washed with brine (30 mL × 2), and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (eluent: 30–60 °C petroleum ether/ethyl acetate = 10/1 to 5/1 to DCM) afforded (*E*)-**4b** as an orange solid; yield: 370.6 mg (100%); mp 239–241 °C (DCM/hexane); ^{1}H NMR (300 MHz, THF-d_8): δ = 11.34 (s, 1H, CH=N), 9.00 (s, 1H, NH), 8.35 (s, 1H, Ar-H), 8.24 (d, J = 8.7 Hz, 1H, Ar-H), 7.65–7.38 (m, 4H, Ar-H), 7.38–7.18 (m, 2H, Ar-H), 6.17 (d, J = 9.9 Hz, 1H, CH=), 3.35–3.08 (m, 1H, CH), 1.26 (d, J = 6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, THF-d_8): δ = 152.7, 146.3, 145.0, 139.7, 136.3, 135.3 (d, J = 2.3 Hz), 131.4, 131.0, 130.8, 129.8, 129.5, 128.7 (q, J = 277.9 Hz), 124.0, 117.9, 40.2 (q, J = 27.9 Hz), 14.9 (d, J = 2.6 Hz); ^{19}F NMR (282 MHz, THF-d_8): δ = −71.7; IR (neat): ν = 3294, 3093, 1618, 1515, 1504, 1338, 1325, 1262, 1188, 1138, 1083, 1017 cm^{-1}; MS (EI, 70 eV): m/z (%) = 408 (M^+, 52.0), 311 (100); elemental analysis calcd. for $C_{18}H_{15}F_3N_4O_4$: C 52.95, H 3.70, N 13.72; found: C 52.67, H 3.81, N 13.72.

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[18] (*E*)-**4b**: $C_{18}H_{15}F_3N_4O_4$, MW = 408.34, triclinic, space group P −1, final *R* indices [I > 2σ(I)], $R1 = 0.0817$, $wR2 = 0.2167$; R indices (all data), $R1 = 0.1102$, $wR2 = 0.2415$; $a = 6.7914$ (5) Å, $b = 7.6553$ (6) Å, $c = 18.2337$ (16) Å, $\alpha = 96.436$ (7)°, $\beta = 99.617$ (7)°, $\gamma = 99.386$ (6)°, $V = 912.63$ (13) Å3, $T = 170$ (2) K, $Z = 2$, reflections collected/unique 5940/3327 ($R_{int} = 0.0235$), number of observations [> 2σ(*I*)] 2380, parameters: 280. CCDC 937724 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.

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